

A Network-Based Approach to Identify the Toxicity Mechanisms of Silver Nanoparticles and Chlorpyrifos in *Danio rerio* (Zebrafish)

**A project report submitted in partial fulfilment
of the requirements for the degree of**

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In

BIOTECHNOLOGY

By

SARANG MAHAJAN

A80404121023

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Under the supervision of

Dr. V. Balasubramanian
Head of the department/ Associate Professor
Amity Institute of Biotechnology

SUBMITTED TO



Amity Institute of Biotechnology
Amity University Chhattisgarh, Raipur

JUNE 2025

DECLARATION

I, the undersigned solemnly declare that the project report entitled “**A Network-Based Approach to Identify the Toxicity Mechanisms of Silver Nanoparticles and Chlorpyrifos in *Danio Rerio* (Zebrafish)**” is conducted in the **Amity Institute of Biotechnology, Amity University Chhattisgarh** as a part of partial fulfilment of the degree. The work carried out during our study under the supervision of **Dr. V. Balasubramanian** is original and conducted by me.

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SARANG MAHAJAN
A80404121023
B.Tech. Biotechnology
Amity Institute of Biotechnology
Amity University Chhattisgarh



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CHHATTISGARH

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Certified that **Sarang Mahajan (A80404121023)** has carried out the research work presented in this project report entitled “**A Network-Based Approach to Identify the Toxicity Mechanisms of Silver Nanoparticles and Chlorpyrifos in *Danio Rerio* (Zebrafish)**” for the award of **Bachelor of Technology in Biotechnology** from Amity University Chhattisgarh, Raipur under my supervision. The project report embodies results of original work, and studies as are carried out by the student himself, and the contents of the report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

SIGNATURE

DR. V. BALASUBRAMANIAN

HEAD OF THE DEPARTMENT

DR. V. BALASUBRAMANIAN
Amity Institute of Biotechnology
Amity University Chhattisgarh

DIRECTOR

PROF. (DR.) VARAPRASAD KOLLA
Amity Institute of Biotechnology
Amity University Chhattisgarh



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SIGNATURE OF SUPERVISOR
DR. V. BALASUBRAMANIAN
Head of the Department/ Associate Professor
Amity Institute of Biotechnology
Amity University Chhattisgarh

Dr. Balasubramanian Velramar

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Date: JUNE 2025
Place: Amity University Chhattisgarh
Biotechnology

SARANG MAHAJAN
A80404121023
B.Tech. BT (VIII)
Amity Institute of
Amity University Chhattisgarh

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ABSTRACT

The increasing presence of engineered nanomaterials and synthetic pesticides in freshwater ecosystems poses a significant environmental concern. Silver nanoparticles (AgNPs) and chlorpyrifos (CPF), a widely used organophosphate pesticide, are among the most commonly detected contaminants. Although both substances exhibit individual toxic effects, their combined and long-term impacts remain insufficiently characterized, particularly in aquatic vertebrate models such as *Danio rerio* (zebrafish). Conventional toxicological studies, often based on chronic exposure, face limitations due to their duration, ethical constraints, and narrow focus, typically ignoring complex biological interactions. In response, computational toxicology offers an advanced, efficient alternative, leveraging multi-scale biological data to explore toxicity mechanisms and predict long-term effects. This study presents a comprehensive computational pipeline for assessing CPF and AgNP toxicity in zebrafish. A network biology framework was employed to identify critical regulatory proteins and disrupted pathways. This was combined with single-cell RNA sequencing (scRNA-seq) to validate gene expression in a tissue-specific context. Molecular docking simulations further elucidated potential interactions with detoxification enzymes, enhancing mechanistic understanding. To quantify toxicity risk, a predictive learning model was developed using chemical, pathway, and gene expression data. This model successfully categorized compounds into low, moderate, or high toxicity groups. *Danio rerio* was selected for its physiological relevance, genetic similarity to humans, and sensitivity to pollutants. The integrated approach described here; combining systems biology, multi-omics, molecular docking, and predictive modelling marks a novel and scalable strategy for evaluating environmental toxicants. This framework not only improves the precision of toxicity prediction but also sets a foundation for assessing other emerging chemical pollutants.

Keywords: *Danio Rerio* (Zebrafish), Silver Nanoparticles, Chlorpyrifos, Network Biology, Transcriptomics, Environmental Toxicology

INTRODUCTION

The exponential rise in the production and environmental dissemination of chemicals, particularly nanomaterials and pesticides, poses a serious challenge to global ecological and human health. With over 100,000 synthetic chemicals registered globally, yet less than 10% thoroughly evaluated for chronic toxicity, there is an urgent need to transition from traditional testing paradigms to mechanism-informed, advanced and rapid systems. (Richard et al., 2016; Thomas et al., 2019).

1. Emergence of Nanomaterials and Pesticides in the Aquatic Environment

Among the most concerning pollutants in aquatic ecosystems are silver nanoparticles (AgNPs) and organophosphate pesticides such as chlorpyrifos (CPF). AgNPs are extensively used in consumer goods, medical devices, and antimicrobial agents. Their widespread usage has led to increasing release into wastewater and sediments, where they interact with aquatic biota and exert multi-system toxic effects (AshaRani et al., 2009; Park et al., 2011).

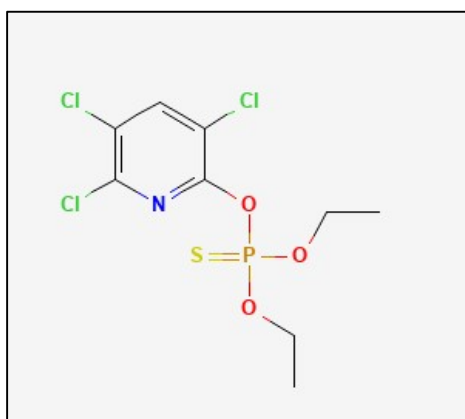


Figure 1: Molecular Structure of Chlorpyrifos ($C_9H_{11}Cl_3NO_3PS$)

Chlorpyrifos, one of the most widely used insecticides globally, is recognized for its neurodevelopmental toxicity, endocrine disruption, and bioaccumulation potential. Despite partial bans, its persistence in soil and water has made it a recurring toxicant in aquatic exposure studies (Slotkin, 2004; Eaton et al., 2008). Organophosphates (OP) are a general class of organic ester compounds containing more than 150 types of phosphorus atoms therein. Due to their strong insecticidal ability and wide range of application, they are widely used around the world as pesticide to control diseases and insect pests, such as dichlorvos and chlorpyrifos, by suppressing the neurotransmitter acetylcholinesterase (AChE).

The toxicological profile of CPF includes neurodevelopmental impairment, oxidative stress, endocrine disruption, and reproductive abnormalities in aquatic species. Sub-lethal exposures in fish embryos have shown alterations in swimming behaviour, gene expression, and brain morphology (Pereira et al., 2013). Because CPF is hydrophobic, it tends to bioaccumulate in lipid-rich tissues of aquatic organisms, potentially magnifying its effects across trophic levels.

Additionally, CPF is often found in environmental mixtures with other pesticides and heavy metals, raising concerns about cumulative and synergistic toxicity. Organophosphates as a group include over 150 structurally similar compounds, many of which operate through shared modes of action, complicating risk assessments (Jaga & Dharmani, 2003). These interactions can lead to enhanced neurotoxic or immunosuppressive outcomes not predicted by single-chemical studies.

Silver nanoparticles are widely deployed over the sectors due to their efficient antimicrobial characteristics. They are found in wound dressings, coatings for medical devices, textiles, food packaging, and personal care products (AshaRani et al., 2009; Park et al., 2011). This broad usage has led to increasing environmental release, predominantly through wastewater treatment effluents, where AgNPs can transform or aggregate depending on pH, ionic strength, and presence of organic matter. Once in aquatic systems, AgNPs interact with dissolved organic compounds, sediments, and biota, often generating silver ions (Ag^+) that contribute significantly to their toxicity (Levard et al., 2012). Aquatic organisms, including algae, daphnia, and fish, have shown evidence of AgNP-related oxidative stress, genotoxicity, and impaired reproduction (Bondarenko et al., 2013).

Environmental studies have detected AgNP residues in river sediments, surface waters, and even drinking water sources. Their nanoscale dimensions allow them to penetrate biological membranes, and their reactivity permits interference with key metabolic and cellular processes. Importantly, their toxicity is not uniform—it varies based on nanoparticle size, coating, aggregation state, and environmental context (Fabrega et al., 2011). Despite growing data, regulatory standards for environmental levels of AgNPs are limited, highlighting a critical gap in nano safety policy.

Together, AgNPs and CPF illustrate the dual threat posed by novel and persistent chemicals in aquatic ecosystems. Their widespread usage, environmental persistence, and multifaceted toxicity demand the application of advanced tools such as transcriptomics, docking, and machine learning for thorough toxicological evaluation. This study addresses this need by

applying a systems toxicology framework to predict long-term risks of these compounds in a zebrafish model, focusing on molecular mechanisms and ecological implications.

Table 1: Pre-studies of metal toxicity on zebrafish

Toxicants	Exposure Conditions	Observed Effects	Mechanisms	References
MgCl ₂	0.5–500 mg/L, up to 96 h, embryonic stage	Disrupted viability, heart and swim bladder anomalies, pigmentation increase, and mesenchymal cell aggregation	Inflammatory response	(Gomes et al., 2023)
Hg	13–401 µg/L, 24–96 h, from embryos	Cardiac swelling, spinal and tail abnormalities, delayed yolk absorption, elevated locomotor activity	Oxidative stress, inhibition of acetylcholinesterase	(Henriques et al., 2023)
Hydroxyquinoline Ru(II) complexes	15–240 µM, 96 h, embryonic exposure	Increased mortality and malformations, disrupted heartbeat and blood flow	Not specified	(M.H. Huang et al., 2023; W. Huang et al., 2023; Y. Huang et al.)
NiCl ₂ , zingerone, rutin	1–10 mg/L, up to 92 h, from embryos	Reduced hatch and survival rates, skeletal deformities, DNA fragmentation, apoptotic activity	Inflammation, oxidative stress	(Köktürk et al., 2023)
Fluoroquinolones, Cd, Cu	0.2 µM, 96 h, embryonic exposure	Genomic damage, increased apoptosis	DNA repair suppression by Cd; fluoroquinolone-DNA interactions	(Jia et al., 2023)
Metal-contaminated water (Al, Cu, Mn, Sr, Zn)	48 h in juveniles, 169 days in adults	Decline in survival among exposed zebrafish	Oxidative stress	(Paylar et al., 2023)
Cd and cordycepin	2 µg/mL, 48–120 h,	Growth abnormalities, apoptotic signals, liver	Oxidative stress and inflammation	(Priya et al., 2023)

	embryonic stage	dysfunction, intestinal inflammation		
Pb, Ni, Cd	50–150 mg/L, various exposures	Tissue alterations in liver and gills, physiological shifts, DNA damage	Predominantly oxidative stress	(Wagh et al., 2023)
Pb and Mn	0.05 mg/L Pb + 0.3 mg/L Mn, 7 days, embryos	Malformation increase, impaired movement behavior	Gut microbiome disruption, ABC transporter modulation	(Xia et al., 2023)
Cu, trichlorfon	2-month-old juveniles, sub-chronic exposure	Hepatic injury, impaired growth, metabolic dysfunction, immunosuppression	Oxidative and immune stress	(Zhang et al., 2023a, b, c, d)

2. Limitations of Conventional Toxicity Testing

Historically, chemical safety has relied on animal-based toxicological assays, such as LD₅₀ determination, acute toxicity in fish and rodents, and chronic exposure studies. While these methodologies have been instrumental in shaping regulatory frameworks, they present several fundamental limitations. Chief among them is their resource-intensive nature—requiring large numbers of animals, extended periods, and significant financial investments. A single chronic rodent bioassay can cost upwards of \$1 million and take more than two years to complete (Hartung, 2009). As the number of industrial chemicals in circulation exceeds 100,000, these methods simply cannot keep pace with the demand for toxicity evaluation (OECD, 2014).

Additionally, traditional assays often fail to capture low-dose or non-monotonic effects and subtle disruptions at the molecular or cellular level. These assays are typically focused on apical endpoints such as mortality, growth inhibition, or reproductive failure, providing limited insight into the mechanistic pathways of toxicity. As a result, they may overlook early biological perturbations or non-lethal outcomes that nonetheless have long-term health consequences. Moreover, these models are often performed in species with significant physiological and genetic differences from humans, leading to translational gaps and potential misclassification of hazards.

Ethical guidelines limit the use of animal models. The 3Rs principle—Replacement, Reduction, and Refinement—has become a cornerstone of responsible animal research, and

many national and international regulations now encourage the development and use of alternative methods wherever possible. Public sentiment and policy pressures have further accelerated the shift away from animal testing, especially for non-essential or cosmetic chemical evaluations (Russell & Burch, 1959; Daneshian et al., 2015).

Perhaps most critically, the conventional testing paradigm does not address the complexity of mixture toxicity or emergent behaviours in biological systems. Real-world exposures often involve simultaneous contact with multiple chemicals, each potentially acting through different or overlapping modes of action. Standard animal tests are poorly suited for evaluating such interactions or for extrapolating across life stages, populations, and species. Moreover, many regulatory studies lack the resolution to detect epigenetic effects, endocrine disruption at environmentally relevant doses, or long-term behavioural alterations.

These scientific and ethical shortcomings have catalysed a global movement toward alternative testing approaches grounded in systems biology and computational toxicology. These include *in vitro* high-throughput screening assays, toxicogenomic profiling, and *in silico* modelling of chemical–biological interactions. International efforts such as the US EPA’s ToxCast program and Europe’s REACH legislation have invested heavily in developing these platforms to accelerate hazard identification and risk prioritization.

In summary, while conventional toxicology has laid the foundation for modern chemical safety assessment, its constraints necessitate the adoption of next-generation tools. These tools not only improve testing efficiency and human relevance but also enhance our mechanistic understanding of toxicity, ultimately contributing to more predictive, ethical, and sustainable regulatory science.

3. *Danio rerio* (Zebrafish) as a Model in Toxicogenomics

The zebrafish (*Danio rerio*) has established itself as a highly versatile and dependable model organism in environmental toxicology and biomedical research. Its unique combination of biological, genetic, and experimental features has led to widespread adoption in studies investigating the toxicological effects of environmental pollutants, pharmaceuticals, and nanomaterials. Among its most notable attributes is the high degree of genetic homology with humans—approximately 70% of human genes have at least one zebrafish ortholog (Howe et

al., 2013). This evolutionary conservation of genetic pathways makes zebrafish particularly suitable for studying conserved toxicological responses and disease mechanisms.

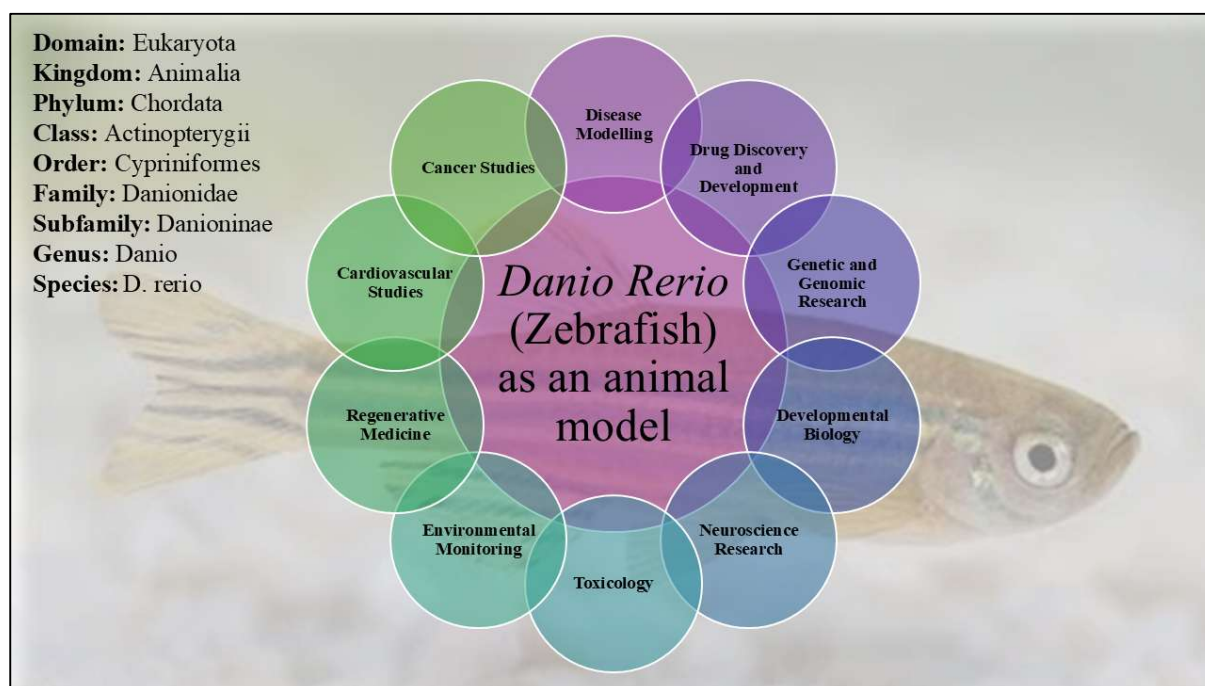


Figure 2: Zebrafish as an animal model

Zebrafish offer several experimental advantages. Their external fertilization and transparent embryos allow direct observation of developmental processes in real time, while their rapid embryogenesis enables the assessment of developmental toxicity within just a few days post-fertilization (Hill et al., 2005). Their small size, high fecundity, and amenability to microplate-based culture systems make them ideal for high-throughput screening applications. Moreover, zebrafish embryos are not considered protected under many animal welfare legislations until 5 days post-fertilization, enabling more flexible study designs under ethical guidelines.

Beyond convenience, zebrafish provide deep mechanistic insights into toxicant action. Their major organ systems—nervous, hepatic, cardiovascular, renal, and immune—develop and function similarly to those of mammals. Zebrafish models have been used to assess neurotoxicity from pesticides like CPF, hepatotoxicity from industrial chemicals, and cardiotoxicity from both xenobiotics and drug candidates (Ali et al., 2011). In particular, their behaviour (e.g., locomotor activity, predator avoidance) can be quantified with high sensitivity to detect subtle neurobehavioral impairments.

With the advent of modern genomic tools, zebrafish have become central to systems-level toxicological studies. Technologies such as bulk RNA-seq, single-cell RNA-seq (scRNA-seq),

transgenesis, and CRISPR/Cas9 gene editing allow researchers to interrogate gene expression changes, manipulate target genes, and track cellular dynamics in vivo. These advances support the construction of adverse outcome pathways (AOPs) by linking molecular initiating events (e.g., receptor binding or oxidative stress) to phenotypic consequences such as malformations or mortality.

Zebrafish are also an excellent platform for multi-omics integration. Proteomics, metabolomics, and epigenomics have been successfully employed in zebrafish to assess the holistic impact of toxicants on biological systems. This capacity is particularly important in evaluating complex substances like AgNPs, where toxicity may arise from multiple mechanisms, including oxidative stress, protein misfolding, or mitochondrial dysfunction.

Importantly, zebrafish models are now being validated and recognized by regulatory authorities. The Organisation for Economic Co-operation and Development (OECD) has set new protocol such as TG 236 (Fish Embryo Acute Toxicity Test), allows the use of zebrafish embryos as a model in testing the chemical hazards. Moreover, large-scale initiatives such as the European Union's Horizon 2020 programs have included zebrafish in their multi-species toxicogenomics platforms.

In conclusion, *Danio rerio* serves as a powerful, scalable, and mechanistically informative model for toxicology research. It bridges the gap between in vitro assays and mammalian in vivo models, offering a cost-effective and ethically viable system that supports the emerging paradigm of predictive, mechanistic toxicology.

4. The Shift to Systems and Network Biology

Traditional toxicology has largely focused on single-molecule interactions and isolated endpoints. However, this reductionist view often fails to capture the emergent properties of biological systems and the multi-target nature of chemical perturbations. Systems biology, and more specifically network biology, addresses this limitation by evaluating how genes, proteins, and metabolites interact within a dynamic and interconnected web. This paradigm shift has enabled a deeper understanding of how environmental toxicants influence not just individual molecules but entire biological networks and pathways (Barabási et al., 2011).

Network biology posits that the function of a biological system arises from the coordinated behaviour of its components. Proteins do not act in isolation—they form complex networks that regulate homeostasis, signal transduction, and cellular function. Perturbation of critical

hubs or bottleneck nodes in these networks by toxicants can lead to disproportionate system-wide effects. For example, the disruption of stress-responsive regulators like TP53 or antioxidant enzymes like SOD1 may not only impair their immediate functions but also propagate instability across DNA repair, apoptosis, and metabolic processes.

Environmental toxicants like silver nanoparticles (AgNPs) and chlorpyrifos (CPF) are especially prone to such network-level perturbations due to their broad and often nonspecific modes of action. AgNPs generate reactive oxygen species (ROS) that interfere with mitochondrial and antioxidant pathways, while CPF inhibits acetylcholinesterase and modulates neurodevelopmental signalling cascades. Network analyses integrating transcriptomic and proteomic data from exposed organisms have shown that these chemicals affect multiple nodes, including transcription factors (e.g., Nrf2), metabolic enzymes (e.g., CYP1A), and signalling intermediates (e.g., MAPKs) (Guney et al., 2016; Yang et al., 2018).

The use of protein-protein interaction (PPI) networks—constructed from curated databases such as STRING, BioGRID, and IntAct—has allowed researchers to visualize how toxicant-responsive genes/proteins are positioned within cellular systems. Centrality analyses (degree, betweenness, closeness) can identify hub proteins, while module detection algorithms (e.g., MCL, Louvain) can uncover functional clusters disrupted by chemical exposure. These modules often correspond to biological processes like xenobiotic metabolism, oxidative stress response, or cell cycle regulation.

Network-based enrichment tools further allow for the contextualization of affected genes within functional ontologies (e.g., GO Biological Process) and canonical pathways (e.g., KEGG, Reactome). For instance, overlaying zebrafish RNA-seq data onto known interactomes has revealed CPF-related disruption in neurotransmitter signalling and AgNP-related impact on mitochondrial homeostasis (Farnsworth et al., 2020). This network-centered perspective provides a holistic and mechanistically interpretable view of toxicity, going beyond binary classification to reveal layered biological responses.

Moreover, network biology supports the generation of hypothesis-driven models such as adverse outcome pathways (AOPs), where molecular initiating events trigger a cascade of key events leading to phenotypic outcomes. By identifying which nodes or pathways are early responders, researchers can prioritize biomarkers for monitoring or targets for therapeutic intervention.

In summary, the integration of systems and network biology into toxicology represents a transformative approach that enhances mechanistic understanding, improves predictive capacity, and supports data-rich regulatory science. It is particularly well-suited to evaluate complex and multifactorial agents like nanomaterials and pesticides, which interact with diverse molecular targets and pathways.

5. RNA-Seq and Single-Cell Transcriptomics in Environmental Toxicology

The advent of RNA sequencing (RNA-seq) has revolutionized the field of toxicology by providing genome-wide, quantitative assessments of gene expression in response to chemical exposure. Unlike traditional gene-by-gene assays, RNA-seq captures dynamic changes across the entire transcriptome, enabling the identification of both known and novel targets of toxicants. This has proven invaluable in elucidating mode-of-action pathways, biomarker discovery, and hazard classification (Wang et al., 2009).

In environmental toxicology, RNA-seq has been applied extensively to model organisms, including zebrafish, to study responses to a wide array of pollutants such as heavy metals, nanoparticles, pesticides, and endocrine disruptors. In zebrafish embryos and larvae, bulk RNA-seq has provided insights into the systemic and tissue-specific impacts of toxicants, revealing coordinated changes in metabolic, inflammatory, and developmental pathways (Marquart et al., 2015).

Single-cell RNA sequencing (scRNA-seq) represents the next frontier by allowing the dissection of heterogeneous tissue responses at the individual cell level. Unlike bulk RNA-seq, which averages signals across thousands of cells, scRNA-seq can resolve subtle yet biologically significant shifts in gene expression within specific subpopulations. This is particularly important in complex tissues like the brain, liver, and kidney, where toxicants may selectively affect one cell type while sparing others.

In zebrafish, scRNA-seq has enabled the mapping of cellular atlases during development and under toxicant exposure. For instance, CPF has been shown to downregulate neurogenesis-related genes in neural progenitor clusters, while AgNP exposure has been linked to mitochondrial stress responses in hepatocyte subtypes (Farnsworth et al., 2020). These studies support a more precise understanding of tissue- and cell-specific mechanisms of action, which is essential for building predictive adverse outcome pathways (AOPs).

Moreover, transcriptomic signatures from RNA-seq studies can be integrated with other omics layers, such as proteomics and metabolomics, to build holistic toxicity profiles. Such multi-omics approaches provide a systems-level understanding of toxicological responses, enabling improved risk stratification and chemical prioritization.

Transcriptomics also contributes to the development of early-response biomarkers. For example, upregulation of oxidative stress genes (e.g., *nrf2a*, *gclc*), apoptosis markers (e.g., *caspase-3*), or immune regulators (e.g., *il1 β*) upon toxicant exposure can serve as sensitive indicators of sub-lethal effects. These gene expression profiles can be leveraged in high-throughput screening platforms to assess environmental chemical libraries.

Advanced computational tools, such as pseudo-time trajectory inference and cell–cell interaction mapping, have further enhanced the utility of scRNA-seq in toxicology. These methods can track the progression of toxicant-induced changes in developmental lineages or identify disruptions in cellular communication networks, such as inflammation or angiogenesis.

In summary, RNA-seq and scRNA-seq offer unparalleled resolution for examining the molecular consequences of toxicant exposure. When applied to model organisms like zebrafish, these technologies support mechanistic toxicology, AOP development, and the refinement of regulatory strategies.

6. Molecular Docking and Target Identification

Molecular docking is a computational technique used to predict the interaction between small molecules (e.g., drugs, pollutants) and their protein targets based on their structural complementarity. As a cornerstone of structure-based drug design, docking has also gained prominence in toxicology, particularly in elucidating molecular initiating events (MIEs) in the adverse outcome pathway (AOP) framework. By estimating the binding affinity and mode of interaction between a chemical and a macromolecular target, docking can identify plausible points of toxicant-induced biological disruption (Trott & Olson, 2010).

The relevance of molecular docking in environmental toxicology lies in its ability to anticipate how chemicals might perturb essential biological processes. For example, chlorpyrifos (CPF), a widely studied organophosphate pesticide, has been docked against a range of zebrafish proteins implicated in detoxification and cellular stress response. High-affinity binding to cytochrome P450 enzymes such as CYP1A suggests potential inhibition of metabolic

detoxification pathways, while interaction with SOD1 (superoxide dismutase) may compromise antioxidant defense. Docking with TP53, a key tumour suppressor involved in apoptosis and DNA repair, implies that CPF may also trigger pro-apoptotic signalling cascades (Bhattacharya et al., 2017).

Such predictions are consistent with experimental data showing oxidative stress, genotoxicity, and apoptosis in CPF-exposed zebrafish embryos and larvae. Thus, molecular docking serves as a powerful hypothesis-generating tool that complements transcriptomic and proteomic approaches by pinpointing initial sites of action. Moreover, it allows prioritization of target proteins for in vitro validation and expands the understanding of toxicity beyond empirical observations.

Silver nanoparticles (AgNPs) pose distinct challenges in toxicity assessment because of their metallic composition and the absence of a clearly defined molecular structure. Traditional ligand–receptor docking paradigms are not directly applicable. However, researchers have begun using small silver cluster models (e.g., Ag₄, Ag₁₃) or silver ion (Ag⁺) interactions to approximate likely binding scenarios. These models suggest that Ag⁺ can interact with cysteine- and histidine-rich regions of proteins, leading to thiol displacement, conformational destabilization, and functional inactivation (Leclerc & Wilkinson, 2014). Docking studies have explored these interactions using metal ion-supported force fields or quantum chemical approximations, particularly in enzymes involved in redox homeostasis or cellular respiration.

Another valuable application of docking in toxicity testing is in identifying off-target interactions. Many environmental chemicals, especially those with lipophilic or reactive moieties, can bind non-selectively to multiple proteins. Screening a compound against a panel of potential zebrafish receptors can reveal unintended or secondary targets, helping to explain phenotypic observations such as developmental delay, neurotoxicity, or metabolic disturbances.

The fusion of docking system to transcriptomics data can deliberately enhance the accuracy. For example, if transcriptomic analysis reveals upregulation of apoptosis-related genes following exposure to a compound, and docking shows strong binding of that compound to TP53 or caspase proteins, the mechanistic link becomes stronger. This multi-evidence approach aligns with systems toxicology principles, improving both predictive power and biological relevance.

In conclusion, molecular docking is a valuable *in silico* method in modern toxicology. It not only predicts molecular initiating events but also aids in understanding complex toxicodynamic behaviour of environmental contaminants. Its scalability, cost-effectiveness, and synergy with omics data make it an indispensable tool in the risk assessment of both conventional pollutants like CPF and emerging materials such as AgNPs.

7. Pathway Enrichment and Functional Interpretation

In modern toxicogenomics, pathway enrichment analysis serves as a critical bridge between large-scale gene expression data and interpretable biological insights. Following differential gene expression analysis (DGEA), enrichment analysis identifies statistically overrepresented biological processes, pathways, or molecular functions among the deregulated genes, thus contextualizing transcriptional changes in terms of cellular behaviour, disease relevance, or toxicity mechanisms.

One of the primary strengths of pathway enrichment is its ability to simplify complex transcriptomic data into coherent biological narratives. Tools such as g:Profiler, DAVID, Enrichr, and Metascape utilize curated databases—Gene Ontology (GO), Kyoto Encyclopaedia of Genes and Genomes (KEGG), Reactome, and WikiPathways—to assign functional meaning to gene lists (Reimand et al., 2016). These tools compute enrichment significance based on hypergeometric or Fisher's exact tests, correcting for multiple hypothesis testing to ensure statistical robustness.

In the context of environmental toxicology, pathway enrichment is especially valuable. Environmental toxicants often produce multifaceted responses, affecting oxidative stress, apoptosis, hormone signalling, DNA repair, and xenobiotic metabolism. For example, in zebrafish exposed to chlorpyrifos (CPF), enriched pathways frequently include neurotransmission, synaptic plasticity, and neuroinflammation, reflecting CPF's neurotoxic profile. Similarly, silver nanoparticles (AgNPs) often enrich pathways involved in oxidative phosphorylation, protein folding, and stress response, aligning with their ROS-generating and proteotoxic effects.

In our study, “pathway hits” were used as a proxy for the extent of biological disruption. This metric reflects the total number of significantly enriched pathways identified per chemical exposure. A high pathway hit count suggests broad transcriptomic perturbation, possibly implicating systemic or multi-organ effects. Conversely, a low pathway hit count indicates a narrower impact, potentially limited to specific cellular processes or tissues. Incorporating this

parameter into machine learning-based toxicity prediction enhances model granularity and aligns computational outputs with mechanistic biology.

Advanced pathway interpretation can further dissect directionality—i.e., whether a pathway is up- or downregulated—and identify regulatory upstream controllers using tools like Ingenuity Pathway Analysis (IPA) or iPathwayGuide. Some platforms also allow network-based enrichment, where co-expression or protein-protein interaction data guide pathway assembly, improving specificity in noisy or heterogeneous datasets.

Functional clustering and redundancy reduction are also important post-enrichment strategies. Gene ontology terms can be highly redundant, with overlapping gene content across terms. Tools such as REVIGO, ClueGO, and g:Profiler's hierarchical filtering help streamline results, grouping functionally related terms into biological themes. This enables clearer biological interpretation and facilitates figure generation for reports or publications.

Ultimately, pathway enrichment is not merely a data-mining step—it is a hypothesis-generating and mechanistic insight tool. It aligns well with the adverse outcome pathway (AOP) concept by highlighting key events (KEs) that connect molecular changes to phenotypic outcomes. When combined with molecular docking, scRNA-seq, and network analysis, pathway enrichment contributes to a comprehensive and systems-level understanding of chemical toxicity.

8. Machine Learning as Predictive Toxicology Scoring Tool

Machine learning (ML) has emerged as a transformative force in predictive toxicology, offering scalable and data-driven methodologies to classify and prioritize chemicals based on complex, multi-dimensional datasets. With the surge in high-throughput screening (HTS), omics technologies, and computational chemistry, ML models can harness thousands of biological and physicochemical features to infer potential toxic effects across various endpoints, including endocrine disruption, hepatotoxicity, carcinogenicity, and neurodevelopmental toxicity (Mayr et al., 2016).

Popular algorithms include Random Forest (RF), support vector machines (SVM), k-nearest neighbors (k-NN), gradient boosting, and deep learning architectures such as artificial neural networks (ANNs). These models have been extensively trained and validated on curated datasets such as ToxCast, Tox21, and REACH databases. They exhibit high accuracy in binary

and multi-class classification tasks, especially when features are biologically informative and properly pre-processed.

The advantage of machine learning in toxicology lies in its ability to recognize complex, non-linear relationships between chemical descriptors and biological outcomes. ML algorithms can accommodate interactions, co-linearity, and higher-order patterns that may not be detectable through traditional statistical analyses. Importantly, supervised learning models rely on labelled data, where outcomes (e.g., toxicity classes) are known, enabling the training of classifiers that generalize to new, unseen chemicals.

In this study, a simplified but transparent ML strategy was implemented using a linear weighted scoring system embedded in Excel. Chemical features such as molecular weight (MW), logP (partition coefficient), LD₅₀ (lethal dose), and pathway hits were normalized and assigned relative weights based on empirical tuning or solver optimization. The model computes a linear combination of these features, producing a continuous predicted value that is then mapped to categorical toxicity labels (low, moderate, high) using calibrated thresholds.

While not as complex as ensemble ML models, this weighted framework offers several strengths. It is interpretable, does not require programming skills or external dependencies, and produces reproducible results. Moreover, it reflects key principles of logistic regression and generalized linear modelling, where coefficients represent the contribution of each feature to the overall toxicity score. This format is ideal for exploratory studies, early-phase screening, or educational purposes in toxicology.

To enhance robustness, the model's predictive accuracy was evaluated using confusion matrices and classification metrics such as accuracy, precision, and recall. Errors in classification were scrutinized to identify patterns of overprediction or underprediction. It was observed that chemicals with moderate pathway disruption but extreme values in LD₅₀ or logP often straddled threshold boundaries, highlighting areas for model refinement.

Future iterations can incorporate additional descriptors (e.g., topological polar surface area, hydrogen bond donors, docking scores) or implement non-linear modelling techniques using open-source ML platforms like Orange, KNIME, or Python's scikit-learn. These platforms also support cross-validation, feature importance ranking, and dimensionality reduction, all of which can improve generalization and reduce overfitting.

Ultimately, integrating ML with systems toxicology and multi-omics data presents a promising path toward ethical, efficient, and mechanistically informed risk assessment. As computational power and biological data availability increase, machine learning will become an indispensable component of next-generation toxicological research.

9. Adverse Outcome Pathways (AOPs): Mechanistic Risk Assessment

The Adverse Outcome Pathway (AOP) framework, championed by the Organisation for Economic Co-operation and Development (OECD), has revolutionized the landscape of regulatory toxicology by providing a structured, mechanistically informed approach to understanding the progression of toxic effects. An AOP is a conceptual construct that links a molecular initiating event (MIE)—such as a chemical binding to a receptor or enzyme—to a cascade of key biological events (KEs) at cellular, tissue, organ, and organismal levels, ultimately leading to an adverse outcome (AO) relevant to risk assessment (Ankley et al., 2010).

The utility of AOPs lies in their ability to integrate diverse types of evidence—from in vitro bioassays, in silico predictions, and omics datasets—into coherent mechanistic narratives. This not only enhances the interpretability of high-throughput screening data but also reduces reliance on animal testing by supporting the development of predictive models and defined approaches. AOPs promote transparency in decision-making and facilitate regulatory harmonization across regions and agencies (Villeneuve et al., 2014).

In the context of environmental toxicants such as chlorpyrifos (CPF), well-characterized AOPs have been established. For CPF, the MIE involves inhibition of acetylcholinesterase (AChE), a key enzyme in neurotransmitter breakdown. This leads to a buildup of acetylcholine in synaptic clefts, triggering a cascade of key events including disrupted neural signalling, impaired neurogenesis, and altered synaptic plasticity. The ultimate AO is often manifested as developmental neurotoxicity, including cognitive and behavioural impairments (Slotkin, 2004; OECD, 2012).

For emerging contaminants like silver nanoparticles (AgNPs), AOP development is more complex due to the multifaceted and dynamic nature of nanoparticle–biological interactions. However, early-stage frameworks have proposed oxidative stress and mitochondrial dysfunction as likely MIEs. These events may trigger inflammatory responses, endoplasmic reticulum stress, and apoptosis, culminating in outcomes such as impaired organ development or immunosuppression (Nel et al., 2009).

The AOP framework also accommodates feedback loops, modulating factors, and species-specific considerations, making it adaptable to varied toxicological contexts. For instance, the same MIE may lead to different AOs depending on life stage, exposure route, or genetic background. This contextual flexibility is particularly valuable in ecological risk assessments involving non-mammalian species such as zebrafish (Knapen et al., 2015).

Importantly, AOPs serve as a foundation for quantitative AOP (qAOP) modelling, wherein key events are parameterized and linked via mathematical relationships. This enables dose–response modelling, sensitivity analysis, and predictive simulation of chemical behaviour under various exposure scenarios. Such applications are invaluable for next-generation risk assessment (NGRA), where efficiency, mechanistic clarity, and ethical compliance are paramount.

In summary, the AOP paradigm bridges molecular biology and regulatory science, empowering toxicologists to translate mechanistic insights into actionable knowledge. It strengthens the credibility of computational models, informs assay development, and supports evidence-based policy decisions in chemical safety.

10. Ethical and Regulatory Imperatives in Computational Toxicology

The ethical landscape of toxicology has undergone a change in basic assumptions in recent decades, driven by increasing public concern over animal welfare, regulatory pressure to reduce vertebrate testing, and the scientific realization that traditional models often lack translational relevance to humans. This has catalysed the evolution of computational toxicology—an interdisciplinary field that combines bioinformatics, systems biology, artificial intelligence, and toxicogenomics to model chemical effects without relying on extensive in vivo experimentation (Kavlock et al., 2012).

At the heart of modern toxicology lies the 3Rs principle—Reduction, Refinement, and Replacement—first introduced by Russell and Burch in 1959. Today, this ethical framework is widely embraced, with regulatory agencies around the world increasingly adopting alternative approaches that reduce or completely avoid the use of animals in testing. Computational approaches align seamlessly with this principle by enabling the development of predictive, mechanism-informed models that rely on in vitro, in silico, and omics-based data (Balls & Combes, 2005).

Regulatory agencies have begun formally integrating computational tools into chemical safety evaluation pipelines. The United States Environmental Protection Agency (US EPA) has established the ToxCast and Tox21 programs, which utilize high-throughput screening (HTS) and computational modelling to prioritize and predict toxicity for thousands of chemicals. Similarly, the European Chemicals Agency (ECHA) has incorporated quantitative structure-activity relationships (QSARs), read-across methods, and adverse outcome pathways (AOPs) into its REACH regulatory framework (OECD, 2019).

At a global level, the OECD has developed comprehensive guidance documents and databases to standardize and validate *in silico* methods, including the OECD QSAR Toolbox and the AOP Knowledge Base. These resources support harmonized risk assessment across countries and foster mutual acceptance of computational data under the OECD Mutual Acceptance of Data (MAD) system (OECD, 2014).

From a scientific standpoint, computational toxicology offers several advantages. It enables the modelling of species-specific effects, accounts for genetic diversity, and supports high-throughput chemical screening, all of which are difficult to achieve with animal models. In addition, these alternative methods tend to be more time-efficient, cost-effective, and better suited to handle complex exposure scenarios, such as chemical mixtures, low-dose effects, and long-term exposures (Hartung, 2009).

Despite these strengths, computational methods face challenges related to data availability, standardization, validation, and interpretability. Regulatory acceptance hinges on the demonstration of model reliability, relevance, and reproducibility. Programs like EU-ToxRisk, PARC (the European Partnership for the Assessment of Risks from Chemicals), and NICEATM (the National Toxicology Program's Interagency Center for the Evaluation of Alternative Toxicological Methods) are working to bridge these gaps by fostering collaboration, advancing data-driven research, and developing scientifically robust testing frameworks.

In summary, the ethical imperative to reduce animal testing, coupled with the regulatory shift toward mechanism-based, high-throughput assessments, has positioned computational toxicology at the forefront of next-generation risk assessment (NGRA). Its continued integration into regulatory practice will depend on sustained investment in method development, validation, and capacity-building across sectors.

LITERATURE REVIEW

- **Silver Nanoparticles (AgNPs): Environmental Behaviour and Toxicity Relevance**

Silver nanoparticles (AgNPs) are increasingly used across a wide range of agricultural as well as industrial and medicinal applications, resulting in their inevitable release into an aquatic ecosystem. This has raised a concern about potential ecological risks, particularly for aquatic organisms like *Danio Rerio* (zebrafish). The environmental behaviour of AgNPs including their transformations in water systems plays a crucial role in determining their biological toxicity.

Recent studies have emphasised that AgNPs undergo various physical, chemical, and biological transformations once introduced to fresh water. These transformations are influenced by both the properties of nanoparticle i.e., intrinsic (surface coating, size, shape) and environmental conditions (pH, ionic strength, presence of organic matter). Such processes can significantly alter the reactivity, bioavailability, and toxicity of AgNPs.

Notably, AgNPs do not act in isolation. Their interaction with coexisting chemical species and natural colloids can either mitigate or exacerbate their toxic effects. Understanding these complex interactions is essential to accurately assess the long-term ecological impacts of AgNP exposure, particularly in sensitive developmental stages of aquatic vertebrates like zebrafish. Despite ongoing research, several knowledge gaps remain especially in understanding the chronic, low-dose effects of AgNPs on biological pathways and gene regulatory networks. These gaps highlight the need for integrative approaches that combine network biology, transcriptomics, and predictive modelling to access their long-term toxicity potential.

In the context of this study, Silver Nanoparticles are investigated not only for their acute toxic potential but also for their influence on molecular pathways within zebrafish. Our multi-omics approach helps bridge the gap between environmental exposure and mechanistic toxicity, providing a more informed basis for risk assessment.

- **Chlorpyrifos: Organophosphate Pesticide**

Pesticides have been essential for improving crop yields by controlling harmful pests. Yet, their widespread and sometimes unregulated application has led to significant environmental challenges, especially affecting aquatic environments. Chlorpyrifos (CPF), a commonly used organophosphate insecticide, is extensively applied in agricultural practices worldwide (Racke, 1993). Its high environmental persistence and chemical mobility have led to its detection in diverse water bodies, including rivers, lakes, estuaries, and even atmospheric deposition via

rainfall (van Dijk et al., 2013). Such contamination not only poses a threat to aquatic biodiversity but also raises significant concerns regarding bioaccumulation, food safety, and potential human health impacts through water and dietary exposure (Anderson et al., 2004). In this review, chlorpyrifos (CPF) was chosen as a focal chemical due to its widespread application in agriculture and its frequent detection in surface waters. We aim to consolidate current knowledge on the toxic effects of CPF on freshwater organisms, with a particular focus on *Danio rerio* (zebrafish). As a well-established model in ecotoxicology, zebrafish offer valuable insights for assessing environmental risks and monitoring the health of aquatic ecosystems.

In aquatic environments, CPF has been shown to inhibit acetylcholinase (AChE) activity, disrupting neural transmission and behaviour in fishes. However, emerging evidence indicates that its toxicological effects extend far beyond AChE inhibition. Studies have reported CPF-induced oxidative stress, apoptosis, and interference with endocrine signalling, highlighting its broader impact on cellular function and gene regulation.

In zebrafish, CPF exposure during early development has been associated with morphological abnormalities, altered locomotion, and disruption of gene expression in pathways related to neurodevelopment, detoxification, and apoptosis. These findings emphasize the need for a more mechanistic understanding of CPF toxicity at the molecular and pathway level.

This study integrates CPF toxicity data using a network biology approach, identifying key disrupted proteins and pathways. By combining RNA-seq-data, protein interaction networks and docking simulations, we aim to uncover CPF's mechanistic footprint in zebrafish and predict long-term toxicity risk using a predictive scoring model.

- **Conventional Toxicity Testing Approaches and Limitations**

Traditional toxicity testing relies heavily on *in vivo* experiments using mammalian models or aquatic organisms to evaluate chemical safety. These methods provide valuable information about acute and chronic effects, they are often constrained by high costs, time intensive protocols, and ethical concerns regarding animal use. Moreover, such approaches typically accesses observable endpoints like mortality, deformities, or behavioural changes, offering limited insights into the underlying molecular mechanisms of toxicity.

This limitation becomes even more significant when evaluating complex substances like silver nanoparticles (AgNPs) or bioactive compounds like chlorpyrifos (CPF), which can trigger subtle but biologically important disruptions in gene expression, protein interactions, and metabolic processes. Conventional methods are generally not designed to detect such network-level or pathway-level specific toxicity.

As a result, there is a need for informative, high throughput and computationally efficient alternatives to traditional testing. In recent years, systems toxicology approaches integrating transcriptomics, protein networks and machine learning have emerged as powerful tools to model and predict chemical effects at multiple biological scales. These methods not only reduce dependency on live-animal testing but also provide predictive insights into long term and low dose toxicity risks.

In this study, we address these limitations by applying multi-omics and computational workflow to predict chronic toxicity and AgNPs and CPF in zebrafish, leveraging bioinformatic tools, molecular docking, and machine learning based scoring.

- **Computational Toxicology as an Alternative**

1. Network Biology Approaches

Network biology has emerged as a vital discipline in systems toxicology, offering a framework to map, analyse, and interpret the complex interactions between genes, proteins, and signalling pathways perturbed by environmental toxicants. Traditional toxicology often views genes or proteins in isolation, but biological functions are inherently modular, and network based. In the context of toxicity assessment, network biology enables the identification of key regulatory nodes (e.g., hub proteins or transcription factors) that mediate chemical-induced responses. Protein-protein interaction (PPI) networks, for instance, can reveal whether a compound preferentially targets central signalling proteins or peripheral effectors, affecting the overall cellular response.

This approach has been particularly successful in elucidating the mechanistic basis of nanoparticle toxicity. For example, exposure to silver nanoparticles (AgNPs) has been shown to trigger redox imbalance and inflammation, which are governed by interconnected networks involving TP53, SOD1, and Nrf2. Tools such as STRING and Cytoscape have been extensively used to model such interactions. Furthermore, network-based enrichment tools like g:Profiler

or Enrichr allow for functional interpretation by linking affected genes or proteins to Gene Ontology (GO) terms and KEGG pathways. These networks not only identify potential biomarkers but also highlight points of convergence between multiple toxicants, which is critical for evaluating combined exposures.

2. RNA-seq Integration

RNA sequencing (RNA-seq), particularly single-cell RNA-seq (scRNA-seq), has revolutionized toxicogenomics by allowing transcriptomic profiling at unprecedented resolution. While conventional RNA-seq offers insights into average expression across tissues, scRNA-seq captures the heterogeneity of cellular responses within complex organs like the brain. This is crucial in developmental and neurotoxicology, where cell-type specificity can influence both sensitivity and outcome.

In zebrafish, scRNA-seq has been used to uncover dynamic gene expression changes during larval development and in response to xenobiotics. Integrating RNA-seq data into network models helps validate whether predicted hub genes or disrupted pathways are transcriptionally active in the tissue of interest. For instance, genes such as *tp53*, *bdnf*, and *sod1*—commonly implicated in oxidative stress and apoptosis—can be confirmed via transcriptomic expression in brain or liver tissues following chemical exposure.

Importantly, RNA-seq also allows the identification of novel targets and unexpected off-target effects, broadening the scope of mechanistic toxicology. Public repositories like NCBI GEO and EMBL-EBI Expression Atlas offer curated datasets that can be reused to validate toxicity predictions across species and contexts.

3. Molecular Docking Simulations

Molecular docking is a cornerstone of structure-based toxicology, offering a predictive method to assess how chemicals interact with target proteins at the atomic level. This is especially relevant when evaluating the inhibitory potential of environmental toxicants on enzymes involved in detoxification, signalling, or metabolism. Docking simulations calculate binding affinities and identify contact residues, helping to infer whether a chemical might interfere with normal protein function.

Tools such as AutoDock Vina, CB-Dock2, and SwissDock have streamlined the docking process, enabling high-throughput virtual screening. These tools have been used to predict interactions between organophosphate pesticides like chlorpyrifos and key metabolic enzymes

such as acetylcholinesterase, CYP450s, and glutathione peroxidases. In the case of zebrafish, docking CPF against orthologous structures of human targets (e.g., TP53, SOD1) provides cross-species insight into conserved toxicological mechanisms.

While molecular docking relies on static insights and doesn't fully reflect the dynamic nature of proteins, it still serves as a valuable initial tool for generating hypotheses. When integrated with gene expression profiles and network analyses, docking helps provide a more functional and integrated perspective in analysing toxicity.

4. Machine Learning as a Toxicity Scoring Tool

Machine learning (ML) models are increasingly being applied to predict chemical toxicity by learning patterns from diverse biological and chemical descriptors. These models, ranging from decision trees to support vector machines and deep neural networks, can integrate multi-dimensional data including molecular structure, physicochemical properties, transcriptomic signatures, and network metrics.

In regulatory toxicology, ML has been used to predict endpoints such as LD50, NOAEL, or toxicity classification (low, moderate, high) using features like molecular weight, logP, and known toxicophores. More recently, integration of omics-derived features, such as pathway disruption scores or docking-based affinity scores, has improved predictive accuracy.

Tools such as Orange, scikit-learn, and even spreadsheet-based solvers allow customizable ML models suited to small- and medium-scale datasets. These models are particularly valuable in contexts where experimental testing is impractical due to ethical or cost constraints. Their ability to handle nonlinear relationships and multi-class classification makes them ideal for environmental toxicity assessments.

- ***Danio rerio* as a Model Organism**

1. Relevance in Toxicogenomics

The zebrafish (*Danio rerio*) has become a premier model organism in developmental biology and toxicology due to its genetic similarity to humans, transparency during early development, and suitability for high-throughput screening. Approximately 70% of human genes have at least one zebrafish ortholog, making it a powerful vertebrate model for studying gene function and chemical responses.

In toxicogenomics, zebrafish are particularly valuable due to the availability of well-annotated genomic resources, including ZFIN and the Zebrafish Genome Project. These databases facilitate mapping of gene expression and phenotype relationships in response to toxicants. Zebrafish embryos can be exposed to chemicals in a controlled aquatic environment, enabling real-time observation of developmental abnormalities, gene expression changes, and survival outcomes.

Their rapid development (from fertilization to organogenesis within 72 hours) and optical transparency allow researchers to directly visualize processes such as neurodevelopment, organogenesis, and circulation. These features make zebrafish a key organism for integrating molecular, phenotypic, and toxicological data.

2. Advantages in Developmental Toxicity Studies

Zebrafish embryos are uniquely suited for studying developmental toxicity due to their external fertilization, synchronous growth, and defined developmental stages. Their use is endorsed by regulatory agencies such as the OECD, which has standardized protocols for fish embryo toxicity testing (FET).

Exposure to chemicals during critical windows of development can result in phenotypic changes including craniofacial defects, tail curvature, pericardial oedema, and impaired locomotion—all of which are quantifiable. Additionally, molecular analyses can be performed in parallel to associate specific phenotypes with gene expression alterations, adding mechanistic depth.

For instance, exposure to chlorpyrifos has been shown to impair zebrafish neural development, reduce AChE activity, and alter the expression of genes involved in axon guidance and synaptic signalling. Silver nanoparticles have similarly been linked to disrupted redox homeostasis, impaired swim behaviour, and upregulation of apoptotic pathways in zebrafish larvae.

Zebrafish thus serve as a bridge between in vitro assays and mammalian models, offering ethical, economic, and biological advantages in developmental and neurotoxicology research.

- **Systems Toxicology Frameworks**

Systems toxicology represents a transformative approach that integrates omics data (genomics, transcriptomics, proteomics, and metabolomics), bioinformatics, and computational modelling to understand the mechanisms underlying chemical-induced perturbations. Unlike traditional toxicology that often relies on endpoint observations, systems toxicology focuses on mapping perturbations across biological networks and cellular pathways.

By modelling the entire system response rather than isolated effects, this approach helps identify adverse outcome pathways (AOPs), potential biomarkers of exposure, and species-specific vulnerabilities. For example, network perturbations caused by CPF and AgNPs can be mapped to specific toxicological endpoints like oxidative stress, apoptosis, or endocrine disruption. The field emphasizes predictive modelling, moving toxicology toward a hypothesis-driven, mechanistic science that can support regulatory risk assessment.

- **Multi-Omics in Predictive Toxicology**

In recent years, multi-omics integration has emerged as a transformative strategy in toxicological sciences, enabling researchers to unravel the complex biological effects of chemical exposures with unprecedented resolution. Unlike traditional approaches that rely on isolated molecular endpoints, multi-omics integrates data from various biological layers—genomics, transcriptomics, proteomics, metabolomics, and epigenomics—to build a holistic, systems-level understanding of toxicity mechanisms.

The rationale behind multi-omics in toxicology lies in the multifactorial nature of toxic responses, which often involve simultaneous changes in gene expression, protein modification, metabolic fluxes, and signalling pathway activities. A single omics layer may reveal perturbation, but integrated data can help infer causality, temporal dynamics, and cellular specificity. For instance, a transcriptomic signal showing oxidative stress may gain stronger predictive value when corroborated by proteomic evidence of mitochondrial dysfunction and metabolomic signatures of lipid peroxidation (Poussin et al., 2014).

In environmental and nanoparticle toxicology, multi-omics approaches are particularly valuable due to the unknown or pleiotropic effects many emerging compounds exert. Silver nanoparticles (AgNPs), for example, are known to affect multiple cellular processes ranging from redox signalling to DNA repair. Transcriptomic profiling may identify upregulation of

stress response genes, but without proteomic validation or metabolic context, these findings may remain inconclusive. The combined analysis of proteomic and metabolomic data alongside gene expression profiles has demonstrated that silver nanoparticles (AgNPs) induce oxidative stress responses while simultaneously impairing the tricarboxylic acid (TCA) cycle and glutathione metabolism. These disruptions lead to significant imbalances in cellular energy production and detoxification mechanisms (AshaRani et al., 2009; Park et al., 2011).

Another important use of multi-omics is in predictive modelling and biomarker discovery. High-throughput platforms such as RNA-seq and LC-MS/MS generate thousands of molecular features per sample. When coupled with supervised machine learning or network inference models, these data can be used to classify chemicals by toxicity class, identify early-response biomarkers, or predict long-term phenotypic outcomes. Projects like Tox21 and ToxCast have embraced this paradigm by combining in vitro screening data with transcriptional and biochemical readouts, enabling better prioritization of environmental chemicals (Richard et al., 2016; Thomas et al., 2019).

Moreover, computational frameworks for omics integration—such as DIABLO, iCluster, and MOFA—have enabled researchers to handle high-dimensional data while preserving inter-omics relationships. These tools allow for unsupervised clustering of exposure profiles, identification of multi-layered biomarkers, and reconstruction of mechanistic networks that span genes to phenotypes. Studies using zebrafish models have revealed that integrating multi-omics data can identify distinct time- and tissue-specific molecular signatures associated with developmental neurotoxicity caused by organophosphates and metal exposure (Tingaud-Sequeira et al., 2013; Lee et al., 2020).

The integration of multi omics include differences in data structure and scale, technical variability across platforms, and the need for robust computational infrastructure and statistical modelling. Additionally, the biological interpretation of integrated signals requires expert domain knowledge and curated databases to map molecular changes onto known pathways or adverse outcome frameworks.

Nevertheless, as tools improve and costs decline, multi-omics approaches are rapidly transitioning from exploratory research to applied toxicology, supporting regulatory frameworks and chemical safety assessment. By capturing the interplay between molecular layers, these integrative methods enhance the biological relevance, mechanistic insight, and predictive power of toxicity studies. In this context, the inclusion of multi-omics information

into machine learning–based toxicity scoring models, as demonstrated in this study, reflects an important step toward more robust, mechanism-driven toxicological evaluation.

- **Ethical and Regulatory Dimensions of Computational Toxicology**

The increasing adoption of computational approaches in toxicology marks not only a scientific advancement but also a major ethical and regulatory transformation. Growing recognition of the limitations associated with traditional animal testing—such as concerns over animal welfare and limited predictive accuracy—has propelled computational toxicology as a promising alternative.

Although animal models have historically been the cornerstone of regulatory toxicology, their application raises significant ethical issues due to the pain, distress, and large number of animals involved, especially in studies of chronic toxicity and development. Additionally, species-specific differences in metabolism, physiology, and molecular pathways often limit how well animal data translate to human health outcomes, potentially complicating regulatory assessments (Hartung, 2009).

In this context, computational toxicology offers an ethical and scientifically progressive path forward. *In silico* tools—including structure–activity relationship models (QSARs), machine learning classifiers, network biology frameworks, and physiologically based pharmacokinetic (PBPK) modelling—enable the prediction of toxicity with minimal or no animal use. Importantly, these methods support rapid and cost-effective screening of thousands of chemicals, many of which remain untested under current regulatory inventories.

Global regulatory agencies have recognized the promise of these methods. For instance, the OECD (Organisation for Economic Co-operation and Development) now endorses validated QSAR models for regulatory submissions under REACH and other frameworks (OECD, 2014). The US EPA has launched major initiatives such as ToxCast and Tox21, which aim to replace animal testing with high-throughput *in vitro* assays and computational models that map chemical activity to adverse outcomes (Thomas et al., 2019).

From an ethical standpoint, the reduction of vertebrate animal use in favour of human-relevant, mechanistic models represents a major step forward. Moreover, it aligns with broader societal

and legislative mandates—such as the European Union’s Cosmetics Directive, which bans animal testing for cosmetic ingredients.

Despite these advances, computational toxicology is not without challenges. Issues of data completeness, model validation, and uncertainty quantification remain. For computational tools to gain full regulatory acceptance, transparency, reproducibility, and robust benchmarking against experimental data are essential. Nevertheless, the trajectory is clear: ethical imperatives and scientific innovation are converging to make computational toxicology a cornerstone of 21st-century safety assessment.

- **Adverse Outcome Pathways (AOPs) in Mechanistic and Regulatory Toxicology**

The Adverse Outcome Pathway (AOP) framework has emerged as a central organizing concept in contemporary toxicology, enabling a structured and mechanistically transparent connection between molecular-level perturbations and population-level adverse outcomes. Developed and promoted by the OECD AOP Development Programme, AOPs support both scientific discovery and regulatory decision-making by systematically linking molecular initiating events (MIEs) to key events (KEs) across biological organization.

AOPs are conceptual constructs rather than compound-specific descriptions, which makes them universally applicable to classes of chemicals that induce similar biological effects. A well-defined AOP typically begins with a defined MIE, such as receptor binding or enzyme inhibition, and progresses through cellular and organ-level events to an adverse outcome such as liver fibrosis, reproductive impairment, or behavioural disruption.

This mechanistic anchoring offers several advantages. First, it facilitates the design of targeted *in vitro* assays that can assess chemical impact on early events, thereby predicting downstream toxicity without the need for animal studies. Second, AOPs enhance the interpretability of complex data from transcriptomics, proteomics, or metabolomics by mapping observed changes to biologically relevant events. Third, AOPs are increasingly used in regulatory frameworks for chemical risk assessment, particularly when experimental data is limited or where animal testing is constrained (Ankley et al., 2010).

In the context of zebrafish toxicology, AOPs have proven particularly valuable. The organism’s genetic tractability, transparent development, and human gene homology enable researchers to map chemical-induced changes across molecular and phenotypic scales. For example,

developmental neurotoxicity AOPs often begin with disruption of neurotransmitter synthesis or oxidative stress, progressing to altered axonal guidance and behavioural changes (Knecht et al., 2017).

Importantly, Adverse Outcome Pathways (AOPs) provide a cohesive framework that brings together multi-omics datasets, molecular docking results, and network biology insights, as illustrated in this study. For example, the identification of TP53, SOD1, and CYP1A as critical targets through network analysis—supported by single-cell RNA sequencing data and docking validation—aligns with known AOPs related to oxidative stress and programmed cell death.

As the AOP knowledgebase continues to expand (e.g., through platforms like AOP-Wiki), these frameworks are expected to underpin next-generation risk assessments, particularly for emerging contaminants, nanoparticles, and mixtures. AOPs also support read-across approaches where data from similar chemicals can inform risk predictions for untested substances.

Ultimately, the adoption of AOPs bridges mechanistic toxicology with regulatory science, enabling a shift from empirical hazard identification to mechanism-based prediction, improving both the speed and accuracy of chemical safety assessments.

OBJECTIVES OF THE STUDY:

- To identify key regulatory proteins and biological pathways disrupted by silver nanoparticles and chlorpyrifos in zebrafish using a network biology approach.
- To integrate RNA-seq transcriptomic data with protein interaction networks for toxicity analysis.
- To perform molecular docking studies to predict the interaction of silver nanoparticles and chlorpyrifos with zebrafish metabolic enzymes.
- To develop a machine learning-based model for toxicity scoring based on long-term risk assessment.

MATERIALS AND METHODOLOGY:

Materials Required:

The study primarily involved in silico (computational) analysis. We employed a range of publicly available databases, computational platforms, and data analysis tools to achieve the multi objective workflow. All components were selected based on their accessibility, relevance to zebrafish toxicology, and compatibility with in-silico methods.

• Databases and Data Sources

1. Comparative Toxicogenomics Database (CTD): Utilized to retrieve curated information on chemical–gene and chemical–pathway interactions specifically for silver nanoparticles and chlorpyrifos.
2. STRING (v11.5): Protein–Protein Interaction (PPI) database used to build interaction networks and assess functional connectivity among target proteins.
3. NCBI Gene Expression Omnibus (GEO): The single-cell RNA sequencing dataset GSE134288 was obtained to analyse gene expression patterns in the zebrafish brain at 6.5 days post-fertilization.
4. ZFIN (Zebrafish Information Network): Provided zebrafish-specific gene function, anatomical expression, and toxicity phenotype references.
5. ToxCast and TG-GATEs: Toxicity databases used to gather physicochemical properties and benchmark toxicity labels for a reference set of chemicals.
6. g:Profiler: Web-based enrichment tool used to perform GO (Gene Ontology) and KEGG pathway enrichment on clustered protein modules.
7. PubChem Compound: Used to download ligand structures (e.g., Chlorpyrifos) in 2D and 3D formats for molecular docking.

• Software and Analytical Platforms

1. Cytoscape (v3.10.3.0): A network visualization platform used to analyse PPI networks.
CytoHubba Plugin: Identified top-ranking hub proteins based on network centrality measures.
MCL Clustering Plugin: Detected functional modules (clusters) within the interaction network.
2. Python (v3.10) and Jupyter Notebook: Used for RNA-seq matrix processing and gene-specific expression analysis.
Key Libraries: Pandas, NumPy, Scanpy, Matplotlib, Seaborn

3. CB-Dock2: A web-based blind docking tool that automatically detects binding cavities and performs molecular docking using AutoDock Vina.
4. Chimera (v1.1.8): Used to visualize protein–ligand docking results and prepare receptor structures. (Alternative tools like Discovery Studio also applicable.)
5. Microsoft Excel (Office 365): Utilized for machine learning model development and toxicity scoring using linear regression with sigmoid activation.
Solver Add-in: Used for model training by minimizing prediction error.
Pivot Tables: Used to generate confusion matrices and calculate classification performance.

- **System Requirements:**

Hardware: Laptop with steady internet access

Operating System: Windows11 Home

Methodology:

This study was designed as a comprehensive in silico analysis to investigate and predict the long-term toxicity of Silver Nanoparticles (AgNPs) and Chlorpyrifos (CPF) in the aquatic model organism *Danio rerio* (zebrafish). The methodology integrated network biology, transcriptomic analysis, molecular docking, and machine learning-based modelling using publicly available tools and databases. The research was executed in four sequential phases, each feeding into the next for cumulative insights.

Phase 1: Network Biology and Functional Enrichment

1.1. Retrieval of Chemical-Associated Genes

Gene interaction data for AgNPs and CPF were retrieved from the Comparative Toxicogenomics Database (CTDbase), which curates chemical-gene associations from literature and experimental studies. Only interactions related to *Danio rerio* were considered.

Procedure:

ctd Illuminating how chemicals affect human health

Comparative Toxicogenomics Database

Home Search About Download Contact Us Help

Chlorpyrifos

Basic Gene Interactions Genes Diseases Phenotypes Comps Pathways GO Exposure Studies Exposure Details References

1-50 of 9,761 results.

Interacting Chemical	Interacting Gene	Interaction	References	Organisms
1. Chlorpyrifos	1700012B07RIKL	[azoxystrobin co-treated with 2-chloro-N-(4-chlorobiphenyl-2-yl)nicotinamide co-treated with Chlorpyrifos co-treated with Glyphosate co-treated with Imidacloprid co-treated with Thiabendazole] results in increased methylation of 1700012B07RIKL gene	1	1
2. Chlorpyrifos	A1CF	Chlorpyrifos results in decreased expression of A1CF mRNA	1	1
3. Chlorpyrifos	A2M	Chlorpyrifos results in increased expression of A2M mRNA	1	1
4. Chlorpyrifos	AAAS	Chlorpyrifos results in decreased expression of AAAS mRNA	1	1
5. Chlorpyrifos-methyl	AACS	Chlorpyrifos-methyl results in decreased expression of AACS mRNA	1	1
6. Chlorpyrifos	AADAT	Chlorpyrifos results in decreased expression of AADAT mRNA	1	1
7. Chlorpyrifos	AAMDC	Chlorpyrifos results in increased expression of AAMDC mRNA	1	1
8. Chlorpyrifos	AAMP	Chlorpyrifos results in increased expression of AAMP mRNA	1	1
9. Chlorpyrifos	AAR2	Chlorpyrifos results in increased expression of AAR2 mRNA	1	1
10. Chlorpyrifos	AARS1	Chlorpyrifos results in decreased expression of AARS1 mRNA	1	1
11. Chlorpyrifos	AASDH	Chlorpyrifos results in decreased expression of AASDH mRNA	1	1
12. Chlorpyrifos	AASS	Chlorpyrifos results in increased expression of AASS mRNA	1	1
13. Chlorpyrifos	AATF	Chlorpyrifos results in decreased expression of AATF mRNA	1	1
14. Chlorpyrifos	AB124611	Chlorpyrifos results in decreased expression of AB124611 mRNA	1	1
15. Chlorpyrifos	ABAT	Chlorpyrifos results in decreased expression of ABAT mRNA	1	1
16. Chlorpyrifos	ABCA1	Chlorpyrifos results in decreased expression of ABCA1 mRNA	1	1
17. Chlorpyrifos	ABCA1	Chlorpyrifos results in increased expression of ABCA1 mRNA	2	2
18. Chlorpyrifos	ABCA17	Chlorpyrifos results in increased expression of ABCA17 mRNA	1	1
19. Chlorpyrifos	ABCA1A	Chlorpyrifos results in decreased expression of ABCA1A mRNA	1	1
20. Chlorpyrifos	ABCA1A	Chlorpyrifos results in increased expression of ABCA1A mRNA	1	1
21. Chlorpyrifos	ABCA1A	[Copper co-treated with Chlorpyrifos] results in increased expression of ABCA1A mRNA	1	1
49. Chlorpyrifos	ABCC2	[[Chlorpyrifos co-treated with decamethrin co-treated with polymarcine co-treated with Glyphosate] results in decreased susceptibility to Cisplatin] which results in increased expression of ABCC2 mRNA	1	1
50. Chlorpyrifos	ABCC2	[[Chlorpyrifos co-treated with decamethrin co-treated with polymarcine co-treated with Glyphosate] results in decreased susceptibility to Fluorouracil] which results in increased expression of ABCC2 mRNA	1	1

1-50 of 9,761 results.

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Figure 3: Interface of CTDbase

1. Go to website: <https://ctdbase.org/>
2. Under keyword search: Chemical Name (e.g., Chlorpyrifos).
3. Filter the Organism Name: *Danio Rerio*.
4. Webpage will land to select “Gene Interaction.”
5. Download the file in .csv format.
6. Follow similar steps for Chemical Name “Silver.”

The dataset was filtered to include only experimentally verified interactions or those supported by mechanistic evidence. This resulted in a non-redundant list of genes potentially affected by each compound.

1.2. Construction of Protein-Protein Interaction (PPI) Networks

1. The list of chemical-associated genes was imported into the STRING v11.5 database (Search Tool for the Retrieval of Interacting Genes/Proteins) to construct zebrafish-specific protein–protein interaction networks.
2. Multiple protein option was checked under search bar and Unique Gene symbol list was entered.
3. The interaction confidence score was set to a minimum of 0.7 to ensure high-confidence associations.

Chemical Name	Gene Symbol	Interaction	Interaction Actions
Silver	ABCC1	[Silver Nitrate results in increased abundance of Silver] which results in increased expression of ABCC1 mRNA	increases^abundanceincreases^expression
Silver	ABCC2	Silver results in increased expression of ABCC2 mRNA	increases^expression
Silver	BAXA	Silver analog results in increased expression of BAXA mRNA	increases^expression
Silver	CAT	Silver analog results in increased expression of CAT mRNA	increases^expression
Silver	CCND1	Silver results in decreased expression of CCND1 mRNA	decreases^expression
Silver	CTNNB1	Silver results in increased expression of CTNNB1 protein	increases^expression
Silver	GSK3B	Silver results in increased expression of GSK3B mRNA	increases^expression
Silver	HSPA5	Silver analog results in increased expression of HSPA5 mRNA	increases^expression
Silver	HSPA5	Silver analog results in increased expression of HSPA5 protein	increases^expression
Silver	MT2	Silver analog affects the expression of MT2 mRNA	affects^expression
Silver	MT2	Silver results in decreased expression of MT2 mRNA	decreases^expression
Silver	MT2	Silver results in increased expression of MT2 mRNA	increases^expression
Silver	PCNA	Silver results in increased expression of PCNA protein	increases^expression
Silver	TP53	Silver analog results in decreased expression of TP53 mRNA	decreases^expression
Chlorpyrifos	ABCA1A	Chlorpyrifos results in increased expression of ABCA1A mRNA	increases^expression
Chlorpyrifos	ABCA1A	[Copper co-treated with Chlorpyrifos] results in increased expression of ABCA1A mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	ABCA5	[Copper co-treated with Chlorpyrifos] results in decreased expression of ABCA5 mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	ABCB5	Chlorpyrifos results in increased expression of ABCB5 mRNA	increases^expression
Chlorpyrifos	ABCC3	Chlorpyrifos results in increased expression of ABCC3 mRNA	increases^expression
Chlorpyrifos	ABCF2	Chlorpyrifos results in increased expression of ABCF2 mRNA	increases^expression
Chlorpyrifos	ABCF2	[Copper co-treated with Chlorpyrifos] results in increased expression of ABCF2 mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	ABHD2B	[Copper co-treated with Chlorpyrifos] results in decreased expression of ABHD2B mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	AB13BP	Chlorpyrifos results in increased expression of AB13BP mRNA	increases^expression
Chlorpyrifos	ACACA	Chlorpyrifos affects the expression of ACACA mRNA	affects^expression
Chlorpyrifos	ACACA	Chlorpyrifos results in decreased expression of ACACA mRNA	decreases^expression
Chlorpyrifos	ACACA	[Copper co-treated with Chlorpyrifos] results in increased expression of ACACA mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	ACACA	[Ethylenethiourea co-treated with Chlorpyrifos] results in increased expression of ACACA mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	ACAT2	Chlorpyrifos results in increased expression of ACAT2 mRNA	increases^expression
Chlorpyrifos	ACAT2	[Copper co-treated with Chlorpyrifos] results in increased expression of ACAT2 mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	ACHE	Chlorpyrifos affects the activity of ACHE protein	affects^activity
Chlorpyrifos	ACHE	Chlorpyrifos results in decreased activity of ACHE protein	decreases^activity
Chlorpyrifos	ACHE	Chlorpyrifos results in decreased expression of ACHE mRNA	decreases^expression
Chlorpyrifos	ACHE	Chlorpyrifos results in decreased expression of ACHE protein	decreases^expression
Chlorpyrifos	ACOX3	Chlorpyrifos results in decreased expression of ACOX3 mRNA	decreases^expression
Chlorpyrifos	ACP5	Chlorpyrifos results in decreased expression of ACP5 mRNA	decreases^expression
Chlorpyrifos	ACTC2	Chlorpyrifos results in increased expression of ACTC2 mRNA	increases^expression
Chlorpyrifos	ACTL6A	[Copper co-treated with Chlorpyrifos] results in increased expression of ACTL6A mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	ADGB	Chlorpyrifos results in decreased expression of ADGB mRNA	decreases^expression
Chlorpyrifos	ADGB	[Copper co-treated with Chlorpyrifos] results in decreased expression of ADGB mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	ADH8A	Chlorpyrifos results in increased expression of ADH8A mRNA	increases^expression
Chlorpyrifos	ADHFE1	[Copper co-treated with Chlorpyrifos] results in decreased expression of ADHFE1 mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	AK4	Chlorpyrifos results in increased expression of AK4 mRNA	increases^expression
Chlorpyrifos	AK4	[Copper co-treated with Chlorpyrifos] results in increased expression of AK4 mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	AKAP9	[Copper co-treated with Chlorpyrifos] results in decreased expression of AKAP9 mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	AMH	Chlorpyrifos results in decreased expression of AMH mRNA	decreases^expression
Chlorpyrifos	AMPD3	[Copper co-treated with Chlorpyrifos] results in increased expression of AMPD3 mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	AMY2A	Chlorpyrifos results in increased expression of AMY2A mRNA	increases^expression
Chlorpyrifos	ANGPTL4	Chlorpyrifos results in decreased expression of ANGPTL4 mRNA	decreases^expression
Chlorpyrifos	ANGPTL4	[Copper co-treated with Chlorpyrifos] results in decreased expression of ANGPTL4 mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	ANXA1B	Chlorpyrifos results in increased expression of ANXA1B mRNA	increases^expression
Chlorpyrifos	ANXA1C	Chlorpyrifos results in decreased expression of ANXA1C mRNA	decreases^expression
Chlorpyrifos	ANXA2A	Chlorpyrifos results in increased expression of ANXA2A mRNA	increases^expression
Chlorpyrifos	APOA2	[Copper co-treated with Chlorpyrifos] results in decreased expression of APOA2 mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	APOC1	[Copper co-treated with Chlorpyrifos] results in increased expression of APOC1 mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	APPA	[Copper co-treated with Chlorpyrifos] results in decreased expression of APPA mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	AQP8	Chlorpyrifos results in increased expression of AQP8 mRNA	increases^expression
Chlorpyrifos	ARG2	Chlorpyrifos results in increased expression of ARG2 mRNA	increases^expression
Chlorpyrifos	ARL5C	Chlorpyrifos results in increased expression of ARL5C mRNA	increases^expression
Chlorpyrifos	ARRDC3A	Chlorpyrifos results in increased expression of ARRDC3A mRNA	increases^expression
Chlorpyrifos	ASCL1A	Chlorpyrifos results in decreased expression of ASCL1A mRNA	decreases^expression
Chlorpyrifos	ATF3	Chlorpyrifos results in increased expression of ATF3 mRNA	increases^expression
Chlorpyrifos	ATP1B1A	[Copper co-treated with Chlorpyrifos] results in decreased expression of ATP1B1A mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	ATP2A1L	[Copper co-treated with Chlorpyrifos] results in decreased expression of ATP2A1L mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	ATXN7L2	[Copper co-treated with Chlorpyrifos] results in increased expression of ATXN7L2 mRNA	affects^cotreatmentincreases^expression
Chlorpyrifos	AZIN1A	[Copper co-treated with Chlorpyrifos] results in decreased expression of AZIN1A mRNA	affects^cotreatmentdecreases^expression
Chlorpyrifos	BAG3	Chlorpyrifos results in increased expression of BAG3 mRNA	increases^expression

[illegible]

Chlorpyrifos	CLDNG	Chlorpyrifos results in decreased expression of CLDNG mRNA	decreases^expression
Chlorpyrifos	CLDNG	[Copper co-treated with Chlorpyrifos] results in decreased expression of CLDNG mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	CLXN	Chlorpyrifos results in decreased expression of CLXN mRNA	decreases^expression
Chlorpyrifos	CNTN2	Chlorpyrifos results in increased expression of CNTN2 mRNA	increases^expression
Chlorpyrifos	COL17A1B	Chlorpyrifos results in decreased expression of COL17A1B mRNA	decreases^expression
Chlorpyrifos	CORT	Chlorpyrifos results in increased expression of CORT mRNA	increases^expression
Chlorpyrifos	COX1	Chlorpyrifos results in decreased expression of COX1 mRNA	decreases^expression
Chlorpyrifos	COX1	Chlorpyrifos results in increased expression of COX1 mRNA	increases^expression
Chlorpyrifos	COX6B1	Chlorpyrifos results in increased expression of COX6B1 mRNA	increases^expression
Chlorpyrifos	CPA5	Chlorpyrifos results in increased expression of CPA5 mRNA	increases^expression
Chlorpyrifos	CPB1	Chlorpyrifos results in increased expression of CPB1 mRNA	increases^expression
Chlorpyrifos	CPT1AA	Chlorpyrifos affects the expression of CPT1AA mRNA	affects^expression
Chlorpyrifos	CPT1AA	Chlorpyrifos results in decreased expression of CPT1AA mRNA	decreases^expression
Chlorpyrifos	CPT1AA	[Ethylenthiourea co-treated with Chlorpyrifos] results in increased expression of CPT1AA mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CRHB	[Chlorpyrifos co-treated with cyhalothrin] results in increased expression of CRHB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CRHB	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] results in increased expression of CRHB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CRHB	[Malathion co-treated with Chlorpyrifos] results in increased expression of CRHB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CSR1P	Chlorpyrifos results in increased expression of CSR1P mRNA	increases^expression
Chlorpyrifos	CST3	Chlorpyrifos results in increased expression of CST3 mRNA	increases^expression
Chlorpyrifos	CTRB1	Chlorpyrifos results in increased expression of CTRB1 mRNA	increases^expression
Chlorpyrifos	CTSB	[Copper co-treated with Chlorpyrifos] results in increased expression of CTSB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CTSC	Chlorpyrifos results in increased expression of CTSC mRNA	increases^expression
Chlorpyrifos	CTSC	[Copper co-treated with Chlorpyrifos] results in increased expression of CTSC mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CTSL1	Chlorpyrifos results in increased expression of CTSL1 mRNA	increases^expression
Chlorpyrifos	CTSZ	[Copper co-treated with Chlorpyrifos] results in increased expression of CTSZ mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CUEDC1A	Chlorpyrifos results in increased expression of CUEDC1A mRNA	increases^expression
Chlorpyrifos	CUL3	Chlorpyrifos results in decreased expression of CUL3 mRNA	decreases^expression
Chlorpyrifos	CXCL18B	[Chlorpyrifos co-treated with cyhalothrin] affects the expression of CXCL18B mRNA	affects^cotreatment affects^expression
Chlorpyrifos	CXCL18B	Chlorpyrifos results in decreased expression of CXCL18B mRNA	decreases^expression
Chlorpyrifos	CXCL18B	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] results in increased expression of CXCL18B mRNA	affects^cotreatment increases^expression
Chlorpyrifos	CYGB1	Chlorpyrifos results in increased expression of CYGB1 mRNA	increases^expression
Chlorpyrifos	CYP11A1.1	Chlorpyrifos results in increased expression of CYP11A1.1 mRNA	increases^expression
Chlorpyrifos	CYP19A1A	[Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of CYP19A1A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	CYP19A1A	Chlorpyrifos results in increased expression of CYP19A1A mRNA	increases^expression
Chlorpyrifos	CYP19A1A	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] affects the expression of CYP19A1A mRNA	affects^cotreatment affects^expression
Chlorpyrifos	CYP1A	Chlorpyrifos inhibits the reaction [beta-Naphthoflavone results in increased activity of CYP1A protein]	decreases^reaction increases^activity
Chlorpyrifos	CYP1A	Chlorpyrifos results in decreased activity of CYP1A protein	decreases^activity
Chlorpyrifos	CYP1A	Chlorpyrifos results in increased expression of CYP1A mRNA	increases^expression
Chlorpyrifos	CYP26A1	Chlorpyrifos results in decreased expression of CYP26A1 mRNA	decreases^expression
Chlorpyrifos	CYP26B1	[Copper co-treated with Chlorpyrifos] results in decreased expression of CYP26B1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	CYP3A65	Chlorpyrifos results in increased expression of CYP3A65 mRNA	increases^expression
Chlorpyrifos	CYP3C1	Chlorpyrifos results in increased expression of CYP3C1 mRNA	increases^expression
Chlorpyrifos	CYP51	Chlorpyrifos results in increased expression of CYP51 mRNA	increases^expression
Chlorpyrifos	CYP8B1.3	Chlorpyrifos results in increased expression of CYP8B1.3 mRNA	increases^expression
Chlorpyrifos	CYT1	Chlorpyrifos results in decreased expression of CYT1 mRNA	decreases^expression
Chlorpyrifos	DAZL	Chlorpyrifos affects the expression of DAZL mRNA	affects^expression
Chlorpyrifos	DCPS	[Copper co-treated with Chlorpyrifos] results in decreased expression of DCPS mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	DEC1R1	Chlorpyrifos results in decreased expression of DEC1R1 mRNA	decreases^expression
Chlorpyrifos	DEC1R1	[Copper co-treated with Chlorpyrifos] results in decreased expression of DEC1R1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	DESMA	Chlorpyrifos results in increased expression of DESMA mRNA	increases^expression
Chlorpyrifos	DGLUCY	[Copper co-treated with Chlorpyrifos] results in increased expression of DGLUCY mRNA	affects^cotreatment increases^expression
Chlorpyrifos	DHCR7	Chlorpyrifos results in increased expression of DHCR7 mRNA	increases^expression
Chlorpyrifos	DHDH.1	[Copper co-treated with Chlorpyrifos] results in increased expression of DHDH.1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	DIO1	Chlorpyrifos affects the expression of DIO1 mRNA	affects^expression
Chlorpyrifos	DIO1	Chlorpyrifos results in increased expression of DIO1 mRNA	increases^expression
Chlorpyrifos	DIO2	Chlorpyrifos affects the expression of DIO2 mRNA	affects^expression
Chlorpyrifos	DIO2	Chlorpyrifos results in decreased expression of DIO2 mRNA	decreases^expression
Chlorpyrifos	DIO2	[Copper co-treated with Chlorpyrifos] results in decreased expression of DIO2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	DIO3A	Chlorpyrifos results in decreased expression of DIO3A mRNA	decreases^expression
Chlorpyrifos	DIO3B	Chlorpyrifos results in decreased expression of DIO3B mRNA	decreases^expression
Chlorpyrifos	DIO3B	Chlorpyrifos results in increased expression of DIO3B mRNA	increases^expression
Chlorpyrifos	DIO3B	[Ethylenthiourea co-treated with Chlorpyrifos] results in increased expression of DIO3B mRNA	affects^cotreatment increases^expression
Chlorpyrifos	DLX4B	Chlorpyrifos results in decreased expression of DLX4B mRNA	decreases^expression
Chlorpyrifos	DNAJA3A	Chlorpyrifos results in increased expression of DNAJA3A mRNA	increases^expression
Chlorpyrifos	DNAJA3A	[Copper co-treated with Chlorpyrifos] results in increased expression of DNAJA3A mRNA	affects^cotreatment increases^expression
Chlorpyrifos	DPP7	Chlorpyrifos results in increased expression of DPP7 mRNA	increases^expression
Chlorpyrifos	DRE-MIR-143-1	Chlorpyrifos results in increased expression of DRE-MIR-143-1 mRNA	increases^expression

Chlorpyrifos	DUS2	Chlorpyrifos results in decreased expression of DUS2 mRNA	decreases^expression
Chlorpyrifos	DUSP2	Chlorpyrifos results in increased expression of DUSP2 mRNA	increases^expression
Chlorpyrifos	DUSP5	Chlorpyrifos results in increased expression of DUSP5 mRNA	increases^expression
Chlorpyrifos	DUT	Chlorpyrifos results in decreased expression of DUT mRNA	decreases^expression
Chlorpyrifos	EBAG9	Chlorpyrifos results in increased expression of EBAG9 mRNA	increases^expression
Chlorpyrifos	EBAG9	[Copper co-treated with Chlorpyrifos] results in increased expression of EBAG9 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	ECH1	Chlorpyrifos results in decreased expression of ECH1 mRNA	decreases^expression
Chlorpyrifos	EC11	Chlorpyrifos results in decreased expression of EC11 mRNA	decreases^expression
Chlorpyrifos	EFHB	Chlorpyrifos results in decreased expression of EFHB mRNA	decreases^expression
Chlorpyrifos	EFHB	[Copper co-treated with Chlorpyrifos] results in decreased expression of EFHB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	EFHD1	Chlorpyrifos results in increased expression of EFHD1 mRNA	increases^expression
Chlorpyrifos	EGR2A	Chlorpyrifos results in increased expression of EGR2A mRNA	increases^expression
Chlorpyrifos	EGR2A	[Copper co-treated with Chlorpyrifos] results in increased expression of EGR2A mRNA	affects^cotreatment increases^expression
Chlorpyrifos	EGR2B	Chlorpyrifos results in increased expression of EGR2B mRNA	increases^expression
Chlorpyrifos	EIF4A1A	Chlorpyrifos results in increased expression of EIF4A1A mRNA	increases^expression
Chlorpyrifos	EIF4EBP3L	Chlorpyrifos results in increased expression of EIF4EBP3L mRNA	increases^expression
Chlorpyrifos	EIF4EBP3L	[Copper co-treated with Chlorpyrifos] results in increased expression of EIF4EBP3L mRNA	affects^cotreatment increases^expression
Chlorpyrifos	ELA2	Chlorpyrifos results in increased expression of ELA2 mRNA	increases^expression
Chlorpyrifos	ELA2L	Chlorpyrifos results in increased expression of ELA2L mRNA	increases^expression
Chlorpyrifos	ELA3L	Chlorpyrifos results in increased expression of ELA3L mRNA	increases^expression
Chlorpyrifos	ELAVL1	Chlorpyrifos results in decreased expression of ELAVL1 mRNA	decreases^expression
Chlorpyrifos	ELOVL6L	Chlorpyrifos results in decreased expression of ELOVL6L mRNA	decreases^expression
Chlorpyrifos	ELOVL6L	[Copper co-treated with Chlorpyrifos] results in decreased expression of ELOVL6L mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ENPP1	[Copper co-treated with Chlorpyrifos] results in increased expression of ENPP1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	ERBIN	[Copper co-treated with Chlorpyrifos] results in decreased expression of ERBIN mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ERF	Chlorpyrifos results in increased expression of ERF mRNA	increases^expression
Chlorpyrifos	ERO1A	Chlorpyrifos results in increased expression of ERO1A mRNA	increases^expression
Chlorpyrifos	ERRF1I	Chlorpyrifos results in increased expression of ERRF1I mRNA	increases^expression
Chlorpyrifos	ESR1	Chlorpyrifos results in decreased expression of ESR1 mRNA	decreases^expression
Chlorpyrifos	ESR1	Chlorpyrifos results in increased expression of ESR1 mRNA	increases^expression
Chlorpyrifos	ESR1	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of ESR1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ESR1	[Malathion co-treated with Chlorpyrifos] results in increased expression of ESR1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	F13A1B	Chlorpyrifos results in increased expression of F13A1B mRNA	increases^expression
Chlorpyrifos	FABP10A	Chlorpyrifos results in decreased expression of FABP10A mRNA	decreases^expression
Chlorpyrifos	FABP10A	[Copper co-treated with Chlorpyrifos] results in decreased expression of FABP10A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	FABP6	Chlorpyrifos results in decreased expression of FABP6 mRNA	decreases^expression
Chlorpyrifos	FAM107B	Chlorpyrifos results in increased expression of FAM107B mRNA	increases^expression
Chlorpyrifos	FASN	Chlorpyrifos results in decreased expression of FASN mRNA	decreases^expression
Chlorpyrifos	FASN	[Ethylene thiourea co-treated with Chlorpyrifos] results in increased expression of FASN mRNA	affects^cotreatment increases^expression
Chlorpyrifos	FBXW11B	Chlorpyrifos results in increased expression of FBXW11B mRNA	increases^expression
Chlorpyrifos	FBXW11B	[Copper co-treated with Chlorpyrifos] results in increased expression of FBXW11B mRNA	affects^cotreatment increases^expression
Chlorpyrifos	FDPS	Chlorpyrifos results in increased expression of FDPS mRNA	increases^expression
Chlorpyrifos	FDPS	[Copper co-treated with Chlorpyrifos] results in increased expression of FDPS mRNA	affects^cotreatment increases^expression
Chlorpyrifos	FLNBA	Chlorpyrifos results in increased expression of FLNBA mRNA	increases^expression
Chlorpyrifos	FLNCA	Chlorpyrifos results in increased expression of FLNCA mRNA	increases^expression
Chlorpyrifos	FLOT1	Chlorpyrifos results in increased expression of FLOT1 mRNA	increases^expression
Chlorpyrifos	FLOT1	[Copper co-treated with Chlorpyrifos] results in increased expression of FLOT1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	FN1	Chlorpyrifos results in increased expression of FN1 mRNA	increases^expression
Chlorpyrifos	FOSAB	Chlorpyrifos results in increased expression of FOSAB mRNA	increases^expression
Chlorpyrifos	FOSL2	Chlorpyrifos results in increased expression of FOSL2 mRNA	increases^expression
Chlorpyrifos	FOXO3A	Chlorpyrifos results in increased expression of FOXO3A protein modified form	increases^expression
Chlorpyrifos	FOXO3A	Chlorpyrifos results in increased phosphorylation of FOXO3A protein	increases^phosphorylation
Chlorpyrifos	FOXO3A	[Ethylene thiourea co-treated with Chlorpyrifos] affects the expression of FOXO3A protein modified form	affects^cotreatment affects^expression
Chlorpyrifos	FOXO3A	[Ethylene thiourea co-treated with Chlorpyrifos] results in decreased expression of FOXO3A protein	affects^cotreatment decreases^expression
Chlorpyrifos	FSIP1	Chlorpyrifos results in decreased expression of FSIP1 mRNA	decreases^expression
Chlorpyrifos	FTR83	Chlorpyrifos results in decreased expression of FTR83 mRNA	decreases^expression
Chlorpyrifos	GADD45A	[Copper co-treated with Chlorpyrifos] results in decreased expression of GADD45A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GADD45B	[Copper co-treated with Chlorpyrifos] results in decreased expression of GADD45B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GALK2	Chlorpyrifos results in decreased expression of GALK2 mRNA	decreases^expression
Chlorpyrifos	GAP43	Chlorpyrifos results in increased expression of GAP43 mRNA	increases^expression
Chlorpyrifos	GAS5	Chlorpyrifos results in decreased expression of GAS5 mRNA	decreases^expression
Chlorpyrifos	GCDHB	[Copper co-treated with Chlorpyrifos] results in decreased expression of GCDHB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GCK	Chlorpyrifos results in decreased expression of GCK mRNA	decreases^expression
Chlorpyrifos	GEMIN4	[Copper co-treated with Chlorpyrifos] results in decreased expression of GEMIN4 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GET3	Chlorpyrifos results in increased expression of GET3 mRNA	increases^expression
Chlorpyrifos	GET3	[Copper co-treated with Chlorpyrifos] results in increased expression of GET3 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	GFAP	Chlorpyrifos results in decreased expression of GFAP mRNA	decreases^expression

Chlorpyrifos	GGCTB	[Copper co-treated with Chlorpyrifos] results in increased expression of GGCTB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	GJA1B	Chlorpyrifos results in decreased expression of GJA1B mRNA	decreases^expression
Chlorpyrifos	GLUD1A	Chlorpyrifos results in increased expression of GLUD1A mRNA	increases^expression
Chlorpyrifos	GMNN	Chlorpyrifos results in decreased expression of GMNN mRNA	decreases^expression
Chlorpyrifos	GMNN	[Copper co-treated with Chlorpyrifos] results in decreased expression of GMNN mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GNB1	[Copper co-treated with Chlorpyrifos] results in decreased expression of GNB1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GNG13B	[Copper co-treated with Chlorpyrifos] results in decreased expression of GNG13B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GNRH3	Chlorpyrifos results in increased expression of GNRH3 mRNA	increases^expression
Chlorpyrifos	GNRH3	[Copper co-treated with Chlorpyrifos] results in increased expression of GNRH3 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	GNSA	Chlorpyrifos results in increased expression of GNSA mRNA	increases^expression
Chlorpyrifos	GPX1A	Chlorpyrifos results in decreased expression of GPX1A mRNA	decreases^expression
Chlorpyrifos	GRNAS	[Copper co-treated with Chlorpyrifos] results in decreased expression of GRNAS mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GSDF	Chlorpyrifos results in decreased expression of GSDF mRNA	decreases^expression
Chlorpyrifos	GSNB	[Copper co-treated with Chlorpyrifos] results in decreased expression of GSNB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GSTP2	Chlorpyrifos results in increased expression of GSTP2 mRNA	increases^expression
Chlorpyrifos	GSTT1B	Chlorpyrifos results in decreased expression of GSTT1B mRNA	decreases^expression
Chlorpyrifos	GSTT1B	[Copper co-treated with Chlorpyrifos] results in decreased expression of GSTT1B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	GTF3AB	[Copper co-treated with Chlorpyrifos] results in decreased expression of GTF3AB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	H1-10	Chlorpyrifos results in increased expression of H1-10 mRNA	increases^expression
Chlorpyrifos	H1M	Chlorpyrifos results in decreased expression of H1M mRNA	decreases^expression
Chlorpyrifos	H1M	[Copper co-treated with Chlorpyrifos] results in decreased expression of H1M mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	HBEGFA	Chlorpyrifos results in increased expression of HBEGFA mRNA	increases^expression
Chlorpyrifos	HDR	[Copper co-treated with Chlorpyrifos] results in decreased expression of HDR mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	HER4.5	Chlorpyrifos results in increased expression of HER4.5 mRNA	increases^expression
Chlorpyrifos	HER6	[Copper co-treated with Chlorpyrifos] results in increased expression of HER6 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	HIF1AL	Chlorpyrifos results in increased expression of HIF1AL mRNA	increases^expression
Chlorpyrifos	HIF1AL	[Copper co-treated with Chlorpyrifos] results in increased expression of HIF1AL mRNA	affects^cotreatment increases^expression
Chlorpyrifos	HIF1AN	Chlorpyrifos results in increased expression of HIF1AN mRNA	increases^expression
Chlorpyrifos	HMGCS1	Chlorpyrifos results in increased expression of HMGCS1 mRNA	increases^expression
Chlorpyrifos	HNRNPK	Chlorpyrifos results in decreased expression of HNRNPK mRNA	decreases^expression
Chlorpyrifos	HNRNPL	[Copper co-treated with Chlorpyrifos] results in increased expression of HNRNPL mRNA	affects^cotreatment increases^expression
Chlorpyrifos	HNRPKL	Chlorpyrifos results in increased expression of HNRPKL mRNA	increases^expression
Chlorpyrifos	HNRPKL	[Copper co-treated with Chlorpyrifos] results in increased expression of HNRPKL mRNA	affects^cotreatment increases^expression
Chlorpyrifos	HP	Chlorpyrifos results in increased expression of HP mRNA	increases^expression
Chlorpyrifos	HRC	[Copper co-treated with Chlorpyrifos] results in decreased expression of HRC mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	HSD17B3	Chlorpyrifos results in increased expression of HSD17B3 mRNA	increases^expression
Chlorpyrifos	HSP70.3	Chlorpyrifos results in decreased expression of HSP70.3 mRNA	decreases^expression
Chlorpyrifos	HSP90AA1.2	Chlorpyrifos results in decreased expression of HSP90AA1.2 mRNA	decreases^expression
Chlorpyrifos	HSPA8	Chlorpyrifos results in increased expression of HSPA8 mRNA	increases^expression
Chlorpyrifos	HSPB11	Chlorpyrifos results in increased expression of HSPB11 mRNA	increases^expression
Chlorpyrifos	HSPB8	Chlorpyrifos results in increased expression of HSPB8 mRNA	increases^expression
Chlorpyrifos	ICN2	Chlorpyrifos results in decreased expression of ICN2 mRNA	decreases^expression
Chlorpyrifos	IFI30	Chlorpyrifos results in decreased expression of IFI30 mRNA	decreases^expression
Chlorpyrifos	IFT57	Chlorpyrifos results in decreased expression of IFT57 mRNA	decreases^expression
Chlorpyrifos	IGF3	Chlorpyrifos affects the expression of IGF3 mRNA	affects^expression
Chlorpyrifos	IGFBP1	Chlorpyrifos results in increased expression of IGFBP1 mRNA	increases^expression
Chlorpyrifos	IGFBP1A	Chlorpyrifos affects the expression of IGFBP1A mRNA	affects^expression
Chlorpyrifos	IGFBP1A	Chlorpyrifos results in decreased expression of IGFBP1A mRNA	decreases^expression
Chlorpyrifos	IGFBP1A	Chlorpyrifos results in increased expression of IGFBP1A mRNA	increases^expression
Chlorpyrifos	IGFBP1A	[Ethylenthionurea co-treated with Chlorpyrifos] affects the expression of IGFBP1A mRNA	affects^cotreatment affects^expression
Chlorpyrifos	IL1B	Chlorpyrifos affects the expression of IL1B mRNA	affects^expression
Chlorpyrifos	IL1B	[Chlorpyrifos co-treated with cyhalothrin] affects the expression of IL1B mRNA	affects^cotreatment affects^expression
Chlorpyrifos	IL1B	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of IL1B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	IL1B	[Malathion co-treated with Chlorpyrifos] results in decreased expression of IL1B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ILK	Chlorpyrifos results in increased expression of ILK mRNA	increases^expression
Chlorpyrifos	ILK	[Copper co-treated with Chlorpyrifos] results in increased expression of ILK mRNA	affects^cotreatment increases^expression
Chlorpyrifos	INAB	Chlorpyrifos results in decreased expression of INAB mRNA	decreases^expression
Chlorpyrifos	INAB	[Copper co-treated with Chlorpyrifos] results in decreased expression of INAB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ING3	Chlorpyrifos results in increased expression of ING3 mRNA	increases^expression
Chlorpyrifos	INHA	Chlorpyrifos results in decreased expression of INHA mRNA	decreases^expression
Chlorpyrifos	INHBB	Chlorpyrifos results in decreased expression of INHBB mRNA	decreases^expression
Chlorpyrifos	INPL1	Chlorpyrifos results in decreased expression of INPL1 mRNA	decreases^expression
Chlorpyrifos	INSIG1	Chlorpyrifos results in increased expression of INSIG1 mRNA	increases^expression
Chlorpyrifos	INSIG1	[Copper co-treated with Chlorpyrifos] results in increased expression of INSIG1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	INSL3	Chlorpyrifos results in increased expression of INSL3 mRNA	increases^expression
Chlorpyrifos	ITM2CB	Chlorpyrifos results in increased expression of ITM2CB mRNA	increases^expression
Chlorpyrifos	ITM2CB	[Copper co-treated with Chlorpyrifos] results in increased expression of ITM2CB mRNA	affects^cotreatment increases^expression

Chlorpyrifos	IVNS1ABPB	Chlorpyrifos results in increased expression of IVNS1ABPB mRNA	increases^expression
Chlorpyrifos	JUNB	Chlorpyrifos results in increased expression of JUNB mRNA	increases^expression
Chlorpyrifos	JUNBB	Chlorpyrifos results in increased expression of JUNBB mRNA	increases^expression
Chlorpyrifos	KDM2BB	[Copper co-treated with Chlorpyrifos] results in increased expression of KDM2BB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	KDR	Chlorpyrifos results in increased expression of KDR mRNA	increases^expression
Chlorpyrifos	KIF11	Chlorpyrifos results in decreased expression of KIF11 mRNA	decreases^expression
Chlorpyrifos	KIF11	[Copper co-treated with Chlorpyrifos] results in decreased expression of KIF11 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	KLF2A	[Copper co-treated with Chlorpyrifos] results in decreased expression of KLF2A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	KLF2B	Chlorpyrifos results in increased expression of KLF2B mRNA	increases^expression
Chlorpyrifos	KRT18	Chlorpyrifos results in increased expression of KRT18 mRNA	increases^expression
Chlorpyrifos	LAD1	Chlorpyrifos results in decreased expression of LAD1 mRNA	decreases^expression
Chlorpyrifos	LAMA5	[Copper co-treated with Chlorpyrifos] results in decreased expression of LAMA5 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	LAPTM4A	Chlorpyrifos results in increased expression of LAPTM4A mRNA	increases^expression
Chlorpyrifos	LDHA	Chlorpyrifos results in increased expression of LDHA mRNA	increases^expression
Chlorpyrifos	LDHA	[Copper co-treated with Chlorpyrifos] results in increased expression of LDHA mRNA	affects^cotreatment increases^expression
Chlorpyrifos	LEF1	Chlorpyrifos results in increased expression of LEF1 mRNA	increases^expression
Chlorpyrifos	LGALS2A	[Copper co-treated with Chlorpyrifos] results in increased expression of LGALS2A mRNA	affects^cotreatment increases^expression
Chlorpyrifos	LGALS3BP.1	[Copper co-treated with Chlorpyrifos] results in increased expression of LGALS3BP.1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	LGMN	[Copper co-treated with Chlorpyrifos] results in increased expression of LGMN mRNA	affects^cotreatment increases^expression
Chlorpyrifos	LIMA1	Chlorpyrifos results in increased expression of LIMA1 mRNA	increases^expression
Chlorpyrifos	LIPG	Chlorpyrifos results in increased expression of LIPG mRNA	increases^expression
Chlorpyrifos	LMBR1L	[Copper co-treated with Chlorpyrifos] results in decreased expression of LMBR1L mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	LONP2	Chlorpyrifos results in increased expression of LONP2 mRNA	increases^expression
Chlorpyrifos	LOXL2B	Chlorpyrifos results in decreased expression of LOXL2B mRNA	decreases^expression
Chlorpyrifos	LOXL2B	[Copper co-treated with Chlorpyrifos] results in decreased expression of LOXL2B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	LPCAT1	[Copper co-treated with Chlorpyrifos] results in decreased expression of LPCAT1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	LPL	Chlorpyrifos results in increased expression of LPL mRNA	increases^expression
Chlorpyrifos	LPL	[Copper co-treated with Chlorpyrifos] results in increased expression of LPL mRNA	affects^cotreatment increases^expression
Chlorpyrifos	LRR8A	[Copper co-treated with Chlorpyrifos] results in increased expression of LRR8A mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MANF	Chlorpyrifos results in decreased expression of MANF mRNA	decreases^expression
Chlorpyrifos	MARCKSA	[Copper co-treated with Chlorpyrifos] results in increased expression of MARCKSA mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MBPA	Chlorpyrifos results in decreased expression of MBPA mRNA	decreases^expression
Chlorpyrifos	MDKB	Chlorpyrifos results in decreased expression of MDKB mRNA	decreases^expression
Chlorpyrifos	MEF2CA	Chlorpyrifos results in decreased expression of MEF2CA mRNA	decreases^expression
Chlorpyrifos	METRNL	Chlorpyrifos results in increased expression of METRNL mRNA	increases^expression
Chlorpyrifos	METRNL	[Copper co-treated with Chlorpyrifos] results in increased expression of METRNL mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MFGE8B	Chlorpyrifos results in decreased expression of MFGE8B mRNA	decreases^expression
Chlorpyrifos	MHC1UDA	Chlorpyrifos results in increased expression of MHC1UDA mRNA	increases^expression
Chlorpyrifos	MHC1UDA	[Copper co-treated with Chlorpyrifos] results in increased expression of MHC1UDA mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MHC1UEA	Chlorpyrifos results in increased expression of MHC1UEA mRNA	increases^expression
Chlorpyrifos	MHC2DAB	Chlorpyrifos results in decreased expression of MHC2DAB mRNA	decreases^expression
Chlorpyrifos	MIBP	Chlorpyrifos results in increased expression of MIBP mRNA	increases^expression
Chlorpyrifos	MIBP	[Copper co-treated with Chlorpyrifos] results in increased expression of MIBP mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MIDN	Chlorpyrifos results in increased expression of MIDN mRNA	increases^expression
Chlorpyrifos	MINDY3	Chlorpyrifos results in increased expression of MINDY3 mRNA	increases^expression
Chlorpyrifos	MIR145	Chlorpyrifos results in increased expression of MIR145 mRNA	increases^expression
Chlorpyrifos	MMP14B	Chlorpyrifos results in increased expression of MMP14B mRNA	increases^expression
Chlorpyrifos	MOB3A	Chlorpyrifos results in increased expression of MOB3A mRNA	increases^expression
Chlorpyrifos	MOB3A	[Copper co-treated with Chlorpyrifos] results in increased expression of MOB3A mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MOXD1	Chlorpyrifos results in increased expression of MOXD1 mRNA	increases^expression
Chlorpyrifos	MPV17	Chlorpyrifos results in decreased expression of MPV17 mRNA	decreases^expression
Chlorpyrifos	MPV17	[Copper co-treated with Chlorpyrifos] results in decreased expression of MPV17 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	MSMO1	Chlorpyrifos results in increased expression of MSMO1 mRNA	increases^expression
Chlorpyrifos	MSMO1	[Copper co-treated with Chlorpyrifos] results in increased expression of MSMO1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MYBPC2B	Chlorpyrifos results in increased expression of MYBPC2B mRNA	increases^expression
Chlorpyrifos	MYCA	Chlorpyrifos results in increased expression of MYCA mRNA	increases^expression
Chlorpyrifos	MYCN	[Copper co-treated with Chlorpyrifos] results in increased expression of MYCN mRNA	affects^cotreatment increases^expression
Chlorpyrifos	MYLK3	[Copper co-treated with Chlorpyrifos] results in decreased expression of MYLK3 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	MYO7A	[Copper co-treated with Chlorpyrifos] results in decreased expression of MYO7A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	MYOZ3A	Chlorpyrifos results in increased expression of MYOZ3A mRNA	increases^expression
Chlorpyrifos	NANOG	Chlorpyrifos results in decreased expression of NANOG mRNA	decreases^expression
Chlorpyrifos	NANOG	[Copper co-treated with Chlorpyrifos] results in decreased expression of NANOG mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NANOS2	Chlorpyrifos results in increased expression of NANOS2 mRNA	increases^expression
Chlorpyrifos	NCALDA	Chlorpyrifos results in increased expression of NCALDA mRNA	increases^expression
Chlorpyrifos	NCAPG	Chlorpyrifos results in decreased expression of NCAPG mRNA	decreases^expression
Chlorpyrifos	NCCRP1	Chlorpyrifos results in increased expression of NCCRP1 mRNA	increases^expression
Chlorpyrifos	NCF1	Chlorpyrifos results in increased expression of NCF1 mRNA	increases^expression

Chlorpyrifos	NDC80	Chlorpyrifos results in decreased expression of NDC80 mRNA	decreases^expression
Chlorpyrifos	NDC80	[Copper co-treated with Chlorpyrifos] results in decreased expression of NDC80 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NDNL2	[Copper co-treated with Chlorpyrifos] results in decreased expression of NDNL2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NDUFA2	Chlorpyrifos results in increased expression of NDUFA2 mRNA	increases^expression
Chlorpyrifos	NDUFB4	[Copper co-treated with Chlorpyrifos] results in decreased expression of NDUFB4 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NEDD4L	Chlorpyrifos results in increased expression of NEDD4L mRNA	increases^expression
Chlorpyrifos	NFE2L2A	[Copper co-treated with Chlorpyrifos] results in decreased expression of NFE2L2A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NFKB1AB	[Copper co-treated with Chlorpyrifos] results in decreased expression of NFKB1AB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NKX2.4B	Chlorpyrifos results in decreased expression of NKX2.4B mRNA	decreases^expression
Chlorpyrifos	NOP56	[Copper co-treated with Chlorpyrifos] results in decreased expression of NOP56 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NPAS4	Chlorpyrifos results in increased expression of NPAS4 mRNA	increases^expression
Chlorpyrifos	NPAS4A	Chlorpyrifos results in decreased expression of NPAS4A mRNA	decreases^expression
Chlorpyrifos	NPC2	[Copper co-treated with Chlorpyrifos] results in increased expression of NPC2 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	NPFFL	Chlorpyrifos results in increased expression of NPFFL mRNA	increases^expression
Chlorpyrifos	NPFFL	[Copper co-treated with Chlorpyrifos] results in increased expression of NPFFL mRNA	affects^cotreatment increases^expression
Chlorpyrifos	NQO1	[Copper co-treated with Chlorpyrifos] results in decreased expression of NQO1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NR1I2	Chlorpyrifos results in increased expression of NR1I2 mRNA	increases^expression
Chlorpyrifos	NRARPB	[Copper co-treated with Chlorpyrifos] results in decreased expression of NRARPB mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	NRIP1B	Chlorpyrifos results in increased expression of NRIP1B mRNA	increases^expression
Chlorpyrifos	NUCKS1A	Chlorpyrifos results in decreased expression of NUCKS1A mRNA	decreases^expression
Chlorpyrifos	OMP	[Copper co-treated with Chlorpyrifos] results in decreased expression of OMP mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	OPRL1	Chlorpyrifos results in increased expression of OPRL1 mRNA	increases^expression
Chlorpyrifos	OR30BT3	[Copper co-treated with Chlorpyrifos] results in decreased expression of OR30BT3 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	OR30BX1	Chlorpyrifos results in decreased expression of OR30BX1 mRNA	decreases^expression
Chlorpyrifos	OR30BX1	[Copper co-treated with Chlorpyrifos] results in decreased expression of OR30BX1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	OR42C4	Chlorpyrifos results in increased expression of OR42C4 mRNA	increases^expression
Chlorpyrifos	OR42E2	[Copper co-treated with Chlorpyrifos] results in decreased expression of OR42E2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	OR70A12	[Copper co-treated with Chlorpyrifos] results in decreased expression of OR70A12 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	OR70A2	[Copper co-treated with Chlorpyrifos] results in decreased expression of OR70A2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ORG	Chlorpyrifos results in decreased expression of ORG mRNA	decreases^expression
Chlorpyrifos	ORG	[Copper co-treated with Chlorpyrifos] results in decreased expression of ORG mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	P4HA1	Chlorpyrifos results in increased expression of P4HA1 mRNA	increases^expression
Chlorpyrifos	PACSLN3	Chlorpyrifos results in decreased expression of PACSLN3 mRNA	decreases^expression
Chlorpyrifos	PAX8	Chlorpyrifos affects the expression of PAX8 mRNA	affects^expression
Chlorpyrifos	PAX9	Chlorpyrifos results in decreased expression of PAX9 mRNA	decreases^expression
Chlorpyrifos	PCK1	Chlorpyrifos results in decreased expression of PCK1 mRNA	decreases^expression
Chlorpyrifos	PCK1	Chlorpyrifos results in increased expression of PCK1 mRNA	increases^expression
Chlorpyrifos	PDE4BA	Chlorpyrifos results in increased expression of PDE4BA mRNA	increases^expression
Chlorpyrifos	PDE4BA	[Copper co-treated with Chlorpyrifos] results in increased expression of PDE4BA mRNA	affects^cotreatment increases^expression
Chlorpyrifos	PDGFRL	Chlorpyrifos results in decreased expression of PDGFRL mRNA	decreases^expression
Chlorpyrifos	PDIA3	[Copper co-treated with Chlorpyrifos] results in increased expression of PDIA3 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	PKD2	Chlorpyrifos results in decreased expression of PKD2 mRNA	decreases^expression
Chlorpyrifos	PKD2	[Copper co-treated with Chlorpyrifos] results in decreased expression of PKD2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	PER2	Chlorpyrifos results in increased expression of PER2 mRNA	increases^expression
Chlorpyrifos	PES	Chlorpyrifos results in increased expression of PES mRNA	increases^expression
Chlorpyrifos	PES	[Copper co-treated with Chlorpyrifos] results in increased expression of PES mRNA	affects^cotreatment increases^expression
Chlorpyrifos	PFDN2	[Copper co-treated with Chlorpyrifos] results in increased expression of PFDN2 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	PFKFB3	Chlorpyrifos results in increased expression of PFKFB3 mRNA	increases^expression
Chlorpyrifos	PFKFB4	Chlorpyrifos results in increased expression of PFKFB4 mRNA	increases^expression
Chlorpyrifos	PGP	Chlorpyrifos results in increased expression of PGP mRNA	increases^expression
Chlorpyrifos	PHLDA2	Chlorpyrifos results in increased expression of PHLDA2 mRNA	increases^expression
Chlorpyrifos	PKLR	Chlorpyrifos results in decreased expression of PKLR mRNA	decreases^expression
Chlorpyrifos	PLK2B	Chlorpyrifos results in increased expression of PLK2B mRNA	increases^expression
Chlorpyrifos	PLK3	Chlorpyrifos results in increased expression of PLK3 mRNA	increases^expression
Chlorpyrifos	PMELB	Chlorpyrifos results in increased expression of PMELB mRNA	increases^expression
Chlorpyrifos	PNP5A	Chlorpyrifos results in increased expression of PNP5A mRNA	increases^expression
Chlorpyrifos	PNP5A	[Copper co-treated with Chlorpyrifos] results in decreased expression of PNP5A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	POLA1	Chlorpyrifos results in decreased expression of POLA1 mRNA	decreases^expression
Chlorpyrifos	POLR1E	Chlorpyrifos results in decreased expression of POLR1E mRNA	decreases^expression
Chlorpyrifos	POU3F2	Chlorpyrifos results in increased expression of POU3F2 mRNA	increases^expression
Chlorpyrifos	POU3F2	[Copper co-treated with Chlorpyrifos] results in increased expression of POU3F2 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	POU5F3	Chlorpyrifos results in increased expression of POU5F3 mRNA	increases^expression
Chlorpyrifos	PPARAB	Chlorpyrifos affects the expression of PPARAB mRNA	affects^expression
Chlorpyrifos	PPARAB	Chlorpyrifos results in decreased expression of PPARAB mRNA	decreases^expression
Chlorpyrifos	PPARG	Chlorpyrifos affects the expression of PPARG mRNA	affects^expression
Chlorpyrifos	PPARG	Chlorpyrifos results in decreased expression of PPARG mRNA	decreases^expression
Chlorpyrifos	PPAT	Chlorpyrifos results in decreased expression of PPAT mRNA	decreases^expression

Chlorpyrifos	PPIA	Chlorpyrifos results in increased expression of PPIA mRNA	increases^expression
Chlorpyrifos	PPP1CAB	Chlorpyrifos results in increased expression of PPP1CAB mRNA	increases^expression
Chlorpyrifos	PPP1CAB	[Copper co-treated with Chlorpyrifos] results in increased expression of PPP1CAB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	PPP1R12A	Chlorpyrifos results in increased expression of PPP1R12A mRNA	increases^expression
Chlorpyrifos	PRDX6	Chlorpyrifos results in increased expression of PRDX6 mRNA	increases^expression
Chlorpyrifos	PRICKLE2	Chlorpyrifos results in increased expression of PRICKLE2 mRNA	increases^expression
Chlorpyrifos	PRR15LA	[Copper co-treated with Chlorpyrifos] results in decreased expression of PRR15LA mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	PSEN1	Chlorpyrifos results in increased expression of PSEN1 mRNA	increases^expression
Chlorpyrifos	PSEN1	[Copper co-treated with Chlorpyrifos] results in increased expression of PSEN1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	PSMB10	Chlorpyrifos results in decreased expression of PSMB10 mRNA	decreases^expression
Chlorpyrifos	PSMB8	Chlorpyrifos results in decreased expression of PSMB8 mRNA	decreases^expression
Chlorpyrifos	PSMB9A	Chlorpyrifos results in decreased expression of PSMB9A mRNA	decreases^expression
Chlorpyrifos	PTGS2A	Chlorpyrifos results in increased expression of PTGS2A mRNA	increases^expression
Chlorpyrifos	PTK2AA	[Copper co-treated with Chlorpyrifos] results in decreased expression of PTK2AA mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	PVALB5	[Copper co-treated with Chlorpyrifos] results in decreased expression of PVALB5 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	PVALB8	[Copper co-treated with Chlorpyrifos] results in decreased expression of PVALB8 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	QDPRB.2	[Copper co-treated with Chlorpyrifos] results in decreased expression of QDPRB.2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	RASGEF1B	Chlorpyrifos results in increased expression of RASGEF1B mRNA	increases^expression
Chlorpyrifos	RDH12L	[Copper co-treated with Chlorpyrifos] results in decreased expression of RDH12L mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	RELT	[Copper co-treated with Chlorpyrifos] results in decreased expression of RELT mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	RFX1B	[Copper co-treated with Chlorpyrifos] results in decreased expression of RFX1B mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	RGS13	Chlorpyrifos results in increased expression of RGS13 mRNA	increases^expression
Chlorpyrifos	RIOK1	Chlorpyrifos results in decreased expression of RIOK1 mRNA	decreases^expression
Chlorpyrifos	RNF32	Chlorpyrifos results in decreased expression of RNF32 mRNA	decreases^expression
Chlorpyrifos	RNF41L	Chlorpyrifos results in decreased expression of RNF41L mRNA	decreases^expression
Chlorpyrifos	RNF41L	[Copper co-treated with Chlorpyrifos] results in decreased expression of RNF41L mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	RPL10	Chlorpyrifos results in decreased expression of RPL10 mRNA	decreases^expression
Chlorpyrifos	RRAD	Chlorpyrifos results in increased expression of RRAD mRNA	increases^expression
Chlorpyrifos	RRAD	[Copper co-treated with Chlorpyrifos] results in increased expression of RRAD mRNA	affects^cotreatment increases^expression
Chlorpyrifos	RSPH14	Chlorpyrifos results in decreased expression of RSPH14 mRNA	decreases^expression
Chlorpyrifos	RUNX2A	Chlorpyrifos results in decreased expression of RUNX2A mRNA	decreases^expression
Chlorpyrifos	S100A10B	Chlorpyrifos results in increased expression of S100A10B mRNA	increases^expression
Chlorpyrifos	S100B	Chlorpyrifos results in increased expression of S100B mRNA	increases^expression
Chlorpyrifos	S100B	[Copper co-treated with Chlorpyrifos] results in increased expression of S100B mRNA	affects^cotreatment increases^expression
Chlorpyrifos	S100Z	[Copper co-treated with Chlorpyrifos] results in decreased expression of S100Z mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	SAA	Chlorpyrifos results in increased expression of SAA mRNA	increases^expression
Chlorpyrifos	SAE2	Chlorpyrifos results in increased expression of SAE2 mRNA	increases^expression
Chlorpyrifos	SCARB1	Chlorpyrifos results in decreased expression of SCARB1 mRNA	decreases^expression
Chlorpyrifos	SCARB1	[Copper co-treated with Chlorpyrifos] results in decreased expression of SCARB1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	SCD	Chlorpyrifos results in increased expression of SCD mRNA	increases^expression
Chlorpyrifos	SEC61B	[Copper co-treated with Chlorpyrifos] results in increased expression of SEC61B mRNA	affects^cotreatment increases^expression
Chlorpyrifos	SEH1L	[Copper co-treated with Chlorpyrifos] results in increased expression of SEH1L mRNA	affects^cotreatment increases^expression
Chlorpyrifos	SELENOT1A	[Copper co-treated with Chlorpyrifos] results in decreased expression of SELENOT1A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	SERPINA1	Chlorpyrifos results in increased expression of SERPINA1 mRNA	increases^expression
Chlorpyrifos	SERPINB1L3	[Copper co-treated with Chlorpyrifos] results in increased expression of SERPINB1L3 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	SFRP5	[Copper co-treated with Chlorpyrifos] results in increased expression of SFRP5 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	SH3BP5LA	Chlorpyrifos results in increased expression of SH3BP5LA mRNA	increases^expression
Chlorpyrifos	SHHA	[Copper co-treated with Chlorpyrifos] results in decreased expression of SHHA mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	SLBP	Chlorpyrifos results in decreased expression of SLBP mRNA	decreases^expression
Chlorpyrifos	SLC13A2	Chlorpyrifos results in increased expression of SLC13A2 mRNA	increases^expression
Chlorpyrifos	SLC16A3	Chlorpyrifos results in increased expression of SLC16A3 mRNA	increases^expression
Chlorpyrifos	SLC22A2	[Copper co-treated with Chlorpyrifos] results in decreased expression of SLC22A2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	SLC25A43	Chlorpyrifos results in increased expression of SLC25A43 mRNA	increases^expression
Chlorpyrifos	SLC2A10	Chlorpyrifos results in decreased expression of SLC2A10 mRNA	decreases^expression
Chlorpyrifos	SLC34A2B	Chlorpyrifos results in increased expression of SLC34A2B mRNA	increases^expression
Chlorpyrifos	SLC3A2B	Chlorpyrifos results in increased expression of SLC3A2B mRNA	increases^expression
Chlorpyrifos	SLC43A2A	Chlorpyrifos results in increased expression of SLC43A2A mRNA	increases^expression
Chlorpyrifos	SLC43A2A	[Copper co-treated with Chlorpyrifos] results in increased expression of SLC43A2A mRNA	affects^cotreatment increases^expression
Chlorpyrifos	SLC44A5B	Chlorpyrifos results in increased expression of SLC44A5B mRNA	increases^expression
Chlorpyrifos	SLC49A4	Chlorpyrifos results in increased expression of SLC49A4 mRNA	increases^expression
Chlorpyrifos	SLC5A5	Chlorpyrifos results in decreased expression of SLC5A5 mRNA	decreases^expression
Chlorpyrifos	SLIT2	Chlorpyrifos results in increased expression of SLIT2 mRNA	increases^expression
Chlorpyrifos	SMAD7	Chlorpyrifos results in increased expression of SMAD7 mRNA	increases^expression
Chlorpyrifos	SNAPIN	Chlorpyrifos results in increased expression of SNAPIN mRNA	increases^expression
Chlorpyrifos	SNRKB	Chlorpyrifos results in increased expression of SNRKB mRNA	increases^expression
Chlorpyrifos	SNRKB	[Copper co-treated with Chlorpyrifos] results in increased expression of SNRKB mRNA	affects^cotreatment increases^expression
Chlorpyrifos	SOCS1	[Copper co-treated with Chlorpyrifos] results in decreased expression of SOCS1 mRNA	affects^cotreatment decreases^expression

Chlorpyrifos	SOCS3A	[Copper co-treated with Chlorpyrifos] results in decreased expression of SOCS3A mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	SOD1	[Malathion co-treated with Chlorpyrifos] results in increased expression of SOD1 mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	SOD2	Chlorpyrifos affects the expression of SOD2 mRNA	affects^expression
Chlorpyrifos	SOD2	Chlorpyrifos results in decreased expression of SOD2 mRNA	decreases^expression
Chlorpyrifos	SOX21A	Chlorpyrifos results in decreased expression of SOX21A mRNA	decreases^expression
Chlorpyrifos	SOX3	Chlorpyrifos results in increased expression of SOX3 mRNA	increases^expression
Chlorpyrifos	SOX3	[Copper co-treated with Chlorpyrifos] results in increased expression of SOX3 mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	SOX4A	Chlorpyrifos results in increased expression of SOX4A mRNA	increases^expression
Chlorpyrifos	SOX4A	[Copper co-treated with Chlorpyrifos] results in increased expression of SOX4A mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	SPAG1	Chlorpyrifos results in increased expression of SPAG1 mRNA	increases^expression
Chlorpyrifos	SPATA20	[Copper co-treated with Chlorpyrifos] results in decreased expression of SPATA20 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	SPRY4	Chlorpyrifos results in increased expression of SPRY4 mRNA	increases^expression
Chlorpyrifos	SPSB4A	Chlorpyrifos results in increased expression of SPSB4A mRNA	increases^expression
Chlorpyrifos	SREBF1	Chlorpyrifos results in decreased expression of SREBF1 mRNA	decreases^expression
Chlorpyrifos	SREBF1	[Ethylene thiourea co-treated with Chlorpyrifos] results in increased expression of SREBF1 protein	affects^cotreatment/increases^expression
Chlorpyrifos	SRGN	Chlorpyrifos results in increased expression of SRGN mRNA	increases^expression
Chlorpyrifos	SRP68	Chlorpyrifos results in decreased expression of SRP68 mRNA	decreases^expression
Chlorpyrifos	SSH2	Chlorpyrifos results in increased expression of SSH2 mRNA	increases^expression
Chlorpyrifos	STAP2	Chlorpyrifos results in increased expression of STAP2 mRNA	increases^expression
Chlorpyrifos	STAP2	[Copper co-treated with Chlorpyrifos] results in increased expression of STAP2 mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	STAR	Chlorpyrifos results in increased expression of STAR mRNA	increases^expression
Chlorpyrifos	SULF2	Chlorpyrifos results in increased expression of SULF2 mRNA	increases^expression
Chlorpyrifos	SULF2	[Copper co-treated with Chlorpyrifos] results in increased expression of SULF2 mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	SULT1ST6	Chlorpyrifos results in decreased expression of SULT1ST6 mRNA	decreases^expression
Chlorpyrifos	SULT2ST1	[Copper co-treated with Chlorpyrifos] results in decreased expression of SULT2ST1 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	SULT2ST2	[Copper co-treated with Chlorpyrifos] results in decreased expression of SULT2ST2 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	SYCP3	Chlorpyrifos results in increased expression of SYCP3 mRNA	increases^expression
Chlorpyrifos	SYN2A	Chlorpyrifos results in decreased expression of SYN2A mRNA	decreases^expression
Chlorpyrifos	SYN2A	Chlorpyrifos results in decreased expression of SYN2A protein	decreases^expression
Chlorpyrifos	TAC3B	[Copper co-treated with Chlorpyrifos] results in increased expression of TAC3B mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	TAT	Chlorpyrifos results in increased expression of TAT mRNA	increases^expression
Chlorpyrifos	TCAP	Chlorpyrifos results in increased expression of TCAP mRNA	increases^expression
Chlorpyrifos	TCF25	Chlorpyrifos results in decreased expression of TCF25 mRNA	decreases^expression
Chlorpyrifos	TENM3	[Copper co-treated with Chlorpyrifos] results in decreased expression of TENM3 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TFPIA	Chlorpyrifos results in increased expression of TFPIA mRNA	increases^expression
Chlorpyrifos	TG	[Ethylene thiourea co-treated with Chlorpyrifos] results in increased expression of TG mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	THRAA	Chlorpyrifos affects the expression of THRAA mRNA	affects^expression
Chlorpyrifos	THRAA	Chlorpyrifos results in increased expression of THRAA mRNA	increases^expression
Chlorpyrifos	THRAA	[Ethylene thiourea co-treated with Chlorpyrifos] results in increased expression of THRAA mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	THRAA	[Malathion co-treated with Chlorpyrifos] results in increased expression of THRAA mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	THRB	Chlorpyrifos affects the expression of THRB mRNA	affects^expression
Chlorpyrifos	THRB	Chlorpyrifos results in decreased expression of THRB mRNA	decreases^expression
Chlorpyrifos	THRB	[Copper co-treated with Chlorpyrifos] results in decreased expression of THRB mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	THRB	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of THRB mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	THY1	Chlorpyrifos results in decreased expression of THY1 mRNA	decreases^expression
Chlorpyrifos	TIMP2B	Chlorpyrifos results in increased expression of TIMP2B mRNA	increases^expression
Chlorpyrifos	TK1	[Copper co-treated with Chlorpyrifos] results in decreased expression of TK1 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TK2	[Copper co-treated with Chlorpyrifos] results in increased expression of TK2 mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	TNFA	Chlorpyrifos affects the expression of TNFA mRNA	affects^expression
Chlorpyrifos	TNFA	[Ethylene thiourea co-treated with Chlorpyrifos] results in increased expression of TNFA mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	TNFAIP8L2B	[Copper co-treated with Chlorpyrifos] results in decreased expression of TNFAIP8L2B mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TNFB	[Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of TNFB mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TNFB	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] affects the expression of TNFB mRNA	affects^cotreatment/affects^expression
Chlorpyrifos	TNFB	[Malathion co-treated with Chlorpyrifos] results in increased expression of TNFB mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	TNNC1	[Copper co-treated with Chlorpyrifos] results in decreased expression of TNNC1 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TNN2C	Chlorpyrifos results in increased expression of TNN2C mRNA	increases^expression
Chlorpyrifos	TOP2A	Chlorpyrifos results in decreased expression of TOP2A mRNA	decreases^expression
Chlorpyrifos	TP53	[Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of TP53 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TP53	[Malathion co-treated with Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of TP53 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TP53	[Malathion co-treated with Chlorpyrifos] results in decreased expression of TP53 mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TP63	Chlorpyrifos results in increased expression of TP63 mRNA	increases^expression
Chlorpyrifos	TPO	Chlorpyrifos results in decreased expression of TPO mRNA	decreases^expression
Chlorpyrifos	TPO	[Ethylene thiourea co-treated with Chlorpyrifos] results in increased expression of TPO mRNA	affects^cotreatment/increases^expression
Chlorpyrifos	TPPP3	Chlorpyrifos results in increased expression of TPPP3 mRNA	increases^expression
Chlorpyrifos	TPTE	Chlorpyrifos results in increased expression of TPTE mRNA	increases^expression
Chlorpyrifos	TRAC	[Copper co-treated with Chlorpyrifos] results in decreased expression of TRAC mRNA	affects^cotreatment/decreases^expression
Chlorpyrifos	TRH	Chlorpyrifos results in increased expression of TRH mRNA	increases^expression

Chlorpyrifos	TRH	[Ethylenethiourea co-treated with Chlorpyrifos] results in increased expression of TRH mRNA	affects^cotreatment increases^expression
Chlorpyrifos	TSC1A	Chlorpyrifos results in decreased expression of TSC1A mRNA	decreases^expression
Chlorpyrifos	TSHBA	[Chlorpyrifos co-treated with cyhalothrin] results in decreased expression of TSHBA mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	TSHBA	Chlorpyrifos results in decreased expression of TSHBA mRNA	decreases^expression
Chlorpyrifos	TSHBA	Chlorpyrifos results in increased expression of TSHBA mRNA	increases^expression
Chlorpyrifos	TSHBA	[Malathion co-treated with Chlorpyrifos] results in increased expression of TSHBA mRNA	affects^cotreatment increases^expression
Chlorpyrifos	TTF1	[Copper co-treated with Chlorpyrifos] results in decreased expression of TTF1 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	TTR	Chlorpyrifos affects the expression of TTR mRNA	affects^expression
Chlorpyrifos	TUBA1B	Chlorpyrifos results in decreased expression of TUBA1B mRNA	decreases^expression
Chlorpyrifos	TUBA1B	Chlorpyrifos results in increased expression of TUBA1B mRNA	increases^expression
Chlorpyrifos	UAP1	Chlorpyrifos results in increased expression of UAP1 mRNA	increases^expression
Chlorpyrifos	UBA2	[Copper co-treated with Chlorpyrifos] results in increased expression of UBA2 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	UCP2	Chlorpyrifos results in decreased expression of UCP2 mRNA	decreases^expression
Chlorpyrifos	UCP2	[Copper co-treated with Chlorpyrifos] results in decreased expression of UCP2 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	UGT1A1	Chlorpyrifos results in increased expression of UGT1A1 mRNA	increases^expression
Chlorpyrifos	UGT1AB	Chlorpyrifos affects the expression of UGT1AB mRNA	affects^expression
Chlorpyrifos	USP7	Chlorpyrifos results in increased expression of USP7 mRNA	increases^expression
Chlorpyrifos	UTS1	Chlorpyrifos results in increased expression of UTS1 mRNA	increases^expression
Chlorpyrifos	UTS1	[Copper co-treated with Chlorpyrifos] results in increased expression of UTS1 mRNA	affects^cotreatment increases^expression
Chlorpyrifos	VANGL2	Chlorpyrifos results in increased expression of VANGL2 mRNA	increases^expression
Chlorpyrifos	VGF	Chlorpyrifos results in increased expression of VGF mRNA	increases^expression
Chlorpyrifos	VSIG10	Chlorpyrifos results in decreased expression of VSIG10 mRNA	decreases^expression
Chlorpyrifos	VSIG10	[Copper co-treated with Chlorpyrifos] results in decreased expression of VSIG10 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	VSIG8A	[Copper co-treated with Chlorpyrifos] results in decreased expression of VSIG8A mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	VTG1	Chlorpyrifos results in increased expression of VTG1 mRNA	increases^expression
Chlorpyrifos	VTG2	Chlorpyrifos results in increased expression of VTG2 mRNA	increases^expression
Chlorpyrifos	VTG4	Chlorpyrifos results in increased expression of VTG4 mRNA	increases^expression
Chlorpyrifos	WDR75	[Copper co-treated with Chlorpyrifos] results in decreased expression of WDR75 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	WDR90	[Copper co-treated with Chlorpyrifos] results in decreased expression of WDR90 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	WNT11	Chlorpyrifos results in decreased expression of WNT11 mRNA	decreases^expression
Chlorpyrifos	XIRP1	Chlorpyrifos results in increased expression of XIRP1 mRNA	increases^expression
Chlorpyrifos	ZFAND2A	Chlorpyrifos results in decreased expression of ZFAND2A mRNA	decreases^expression
Chlorpyrifos	ZNF346	Chlorpyrifos results in increased expression of ZNF346 mRNA	increases^expression
Chlorpyrifos	ZP2.2	Chlorpyrifos results in decreased expression of ZP2.2 mRNA	decreases^expression
Chlorpyrifos	ZP3	Chlorpyrifos results in decreased expression of ZP3 mRNA	decreases^expression
Chlorpyrifos	ZP3	[Copper co-treated with Chlorpyrifos] results in decreased expression of ZP3 mRNA	affects^cotreatment decreases^expression
Chlorpyrifos	ZP3A.1	Chlorpyrifos results in decreased expression of ZP3A.1 mRNA	decreases^expression
Chlorpyrifos	ZP3A.1	[Copper co-treated with Chlorpyrifos] results in decreased expression of ZP3A.1 mRNA	affects^cotreatment decreases^expression

4. Disconnected nodes and isolated interactions were excluded to retain network integrity. The resulting network was downloaded in a tabular format for further analysis.

1.3. Visualization and Analysis in Cytoscape

1. Cytoscape v3.10.3.0 was used to visualize the PPI network.
2. Nodes represented proteins, while edges denoted functional or physical interactions. Node size and color were mapped to interaction degree and clustering, respectively.
3. Hub proteins were identified using the CytoHubba plugin, based on centrality metrics such as Degree, Betweenness, and MCC (Maximal Clique Centrality).
4. Highly ranked hub proteins included tp53, sod1, gpx1a, bdnf, and cyp1a.
5. A raw network file was downloaded, which appeared as:

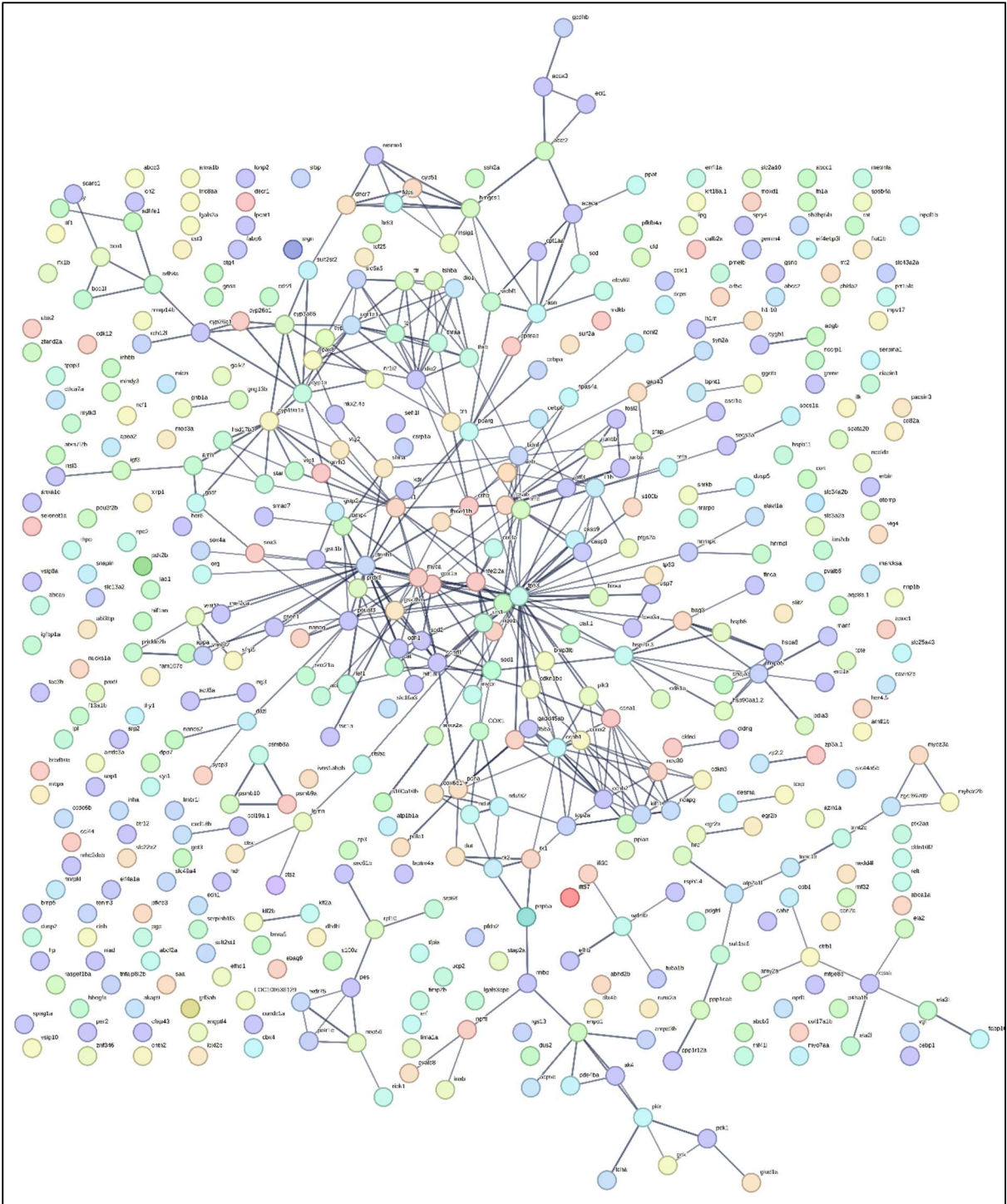


Figure 4: Raw clusters obtained on Cytoscape analysis indicating gene networks

1.4. Clustering and Functional Module Detection

To obtain functionally related protein groups, the network was analysed using the Markov Cluster Algorithm (MCL) through the clusterMaker2 add-in. The resulting clusters were considered as potential biological modules, and each was exported for subsequent functional enrichment analysis.

1.5. Gene Ontology and KEGG Pathway Enrichment

The g:Profiler tool was employed to perform enrichment analysis for each cluster. Overrepresented Gene Ontology Biological Process (GO:BP) terms and KEGG pathways were identified using an FDR-adjusted p-value cutoff of 0.05. Enrichment scores and intersection genes were recorded. The number of significantly affected pathways per chemical was tabulated and later used as a toxicity feature.

Phase 2: Integration of RNA-Seq Transcriptomics

2.1. Data Acquisition and Overview

Single-cell RNA-seq (scRNA-seq) data were downloaded from the NCBI GEO database under the accession number GSE134288. This dataset contained ~6,500 zebrafish brain cells collected from transgenic *nr1d1:VNP* larvae at 6.5 days post-fertilization, generated using the 10X Genomics Chromium platform.

2.2. Data Preprocessing and Filtering

The gene-cell expression matrix was loaded and pre-processed using Python (v3.10) and the Scanpy library. Quality control filters were applied to retain cells with >500 genes expressed, and genes detected in >3 cells. Mitochondrial genes and low-abundance transcripts were removed. Library-size normalization and log_{1p} transformation were applied.

Following code was followed:

```

C > Users > saran > OneDrive > Desktop > filtered_scRNAseq_matrix.csv
1  import pandas as pd
2  import numpy as np
3  import scanpy as sc
4  import matplotlib.pyplot as plt
5  import seaborn as sns
6
7  expression_data = pd.read_csv("GSM3941431_gene_cell_matrix.txt", sep="\t", index_col=0)
8
9  print("Matrix shape:", expression_data.shape) # Genes x Cells
10 print("First 5 rows of the data:")
11 print(expression_data.head())
12
13 min_genes_per_cell = 500
14 cell_sums = (expression_data > 0).sum(axis=0)
15 filtered_cells = cell_sums[cell_sums > min_genes_per_cell].index
16 filtered_data = expression_data[filtered_cells]
17
18 print("\nFiltered matrix shape:", filtered_data.shape)
19
20 normalized_data = np.log1p(filtered_data)
21
22 genes_of_interest = ['tp53', 'sod1', 'bcl2a', 'bdnf', 'gpx1a', 'ccnb1', 'ccnd1', 'ctnnb1', 'ugt1a1', 'esr1']
23
24 normalized_data.index = normalized_data.index.str.lower()
25 genes_of_interest = [g.lower() for g in genes_of_interest]
26
27 summary_results = {}
28 for gene in genes_of_interest:
29     if gene in normalized_data.index:
30         expr = normalized_data.loc[gene]
31         summary = expr.describe()
32         summary_results[gene.upper()] = summary
33         print(f"\n {gene.upper()} expression summary:\n{summary}")
34     else:
35         print(f" {gene.upper()} not found in matrix.")
36
37 pd.DataFrame(summary_results).T.to_csv("gene_expression_summaries.csv")
38
39
40

```

Output file was saved as: gene_expression_summaries.csv as expression table was obtained.

2.3. Expression Profiling of Target Genes

Genes identified during the network analysis phase (including *tp53*, *sod1*, *gpx1a*, *ccnb1*, and *bcl2a*) were extracted from the filtered expression matrix. Descriptive statistics such as mean, median, and maximum expression levels were calculated for these genes. Their expression patterns were then compared across different cell clusters to confirm the transcriptional activity of toxicity-related genes within the zebrafish brain.

2.4. Integration with Network Data

Expression profiles were visually overlaid on the Cytoscape network using expression-based node colouring. This confirmed that hub proteins were not only central but also transcriptionally active in the zebrafish brain at the tested developmental stage, reinforcing their biological relevance.

Following colour pattern was assigned:

Colour Pattern

Bright Red

Orange to Yellow

Blue

Expression Pattern

Highly expressed in brain baseline

Moderately expresses in brain baseline

Low or no expression

Phase 3: Molecular Docking Simulations**3.1. Ligand and Protein Structure Preparation**

1. Chlorpyrifos 2D and 3D structures were retrieved from PubChem (CID: 2730) and converted to docking-ready PDB format using Open Babel.
2. Receptor proteins including tp53, sod1, cyp1a, gpx1a, and bdnf were obtained from the Protein Data Bank (PDB), prioritizing zebrafish homologs when available.
3. Non-standard residues and water molecules were removed. Hydrogen atoms were added.

3.2. Docking Execution with CB-Dock2

Molecular docking was performed using CB-Dock2, a web-based blind docking platform. Receptors were uploaded individually and docked with Chlorpyrifos. The tool auto-identified potential binding cavities and used AutoDock Vina to evaluate binding affinity. Five top-ranked cavities were reported per protein, and the best-scoring conformation was selected.

3.3. Results Extraction and Scoring

Docking scores (in kcal/mol) and cavity volumes were recorded. Binding affinities ranged from -4.2 to -6.2 kcal/mol across different protein targets. These scores were used as proxy indicators of molecular disruption potential and recorded as features for the machine learning model.

AgNPs could not be docked due to structural limitations; thus, toxicity predictions for AgNPs relied on PPI and expression data.

Phase 4: Machine Learning-Based Toxicity Prediction

4.1. Dataset Compilation

A structured dataset of 18 chemicals was compiled manually in Microsoft Excel. Each entry included molecular weight, logP, LD50 (mg/kg), number of pathway hits (from Phase 1), and known toxicity label (Low=0, Moderate=1, High=2).

Data were sourced from CTD, ZFIN, TG-GATEs, and peer-reviewed publications.

Table 2: Raw dataset prepared using database and publications

Chemical Name	Mw	log P	LD50 (mg/kg)	Pathway hits	Toxicity Encoded	References
Chlorpyrifos	350.6	4.7	82	12	2	PubChem CID 2730
AgNP	108	0	0	9	1	PubChem CID 23954
Cadmium	112.4	0	6.1	10	2	PubChem CID 23973
Bisphenol A (BPA)	228.3	3.3	325	7	1	PubChem CID 6623
Ethanol	46.1	-0.3	7060	2	0	PubChem CID 702
Atrazine	215.7	2.6	3090	9	1.5	PubChem CID 2256
Lead (Pb ²⁺)	207.2	0	30	10	2	PubChem CID 5352425
Triclosan	289.5	4.8	3700	7	1.5	PubChem CID 5564
TCDD (Dioxin)	322	6.8	0.001	65	2	PubChem CID 15625

Diazinon	304.4	3.8	696	500	2	PubChem CID 3017
Acetaminophen	151.2	0.9	388	4	0.5	PubChem CID 1983
Ibuprofen	206.3	3.5	636	3	0	PubChem CID 3672
Fluoxetine	309.3	4.1	284	5	0.5	PubChem CID 3386
Caffeine	194.2	-0.1	192	4	0.5	PubChem CID 2519
Nicotine	162.2	1.2	50	15	1.5	PubChem CID 89594
Permethrin	391.3	6.1	430	12	1	PubChem CID 40326
Glyphosate	169.1	-3.2	5600	6	0.5	PubChem CID 3496
Copper (Cu ²⁺)	63.5	0	3.5	8	1.5	PubChem CID 23978

4.2. Feature Engineering and Scaling

1. All numeric features were normalized using min-max scaling.
2. Columns for MW_scaled, logP_scaled, LD50_scaled, and pathway_scaled were created and computed using the formula:

$$\text{Scaled Value} = \frac{(\text{Value} - \text{Min Value})}{(\text{Max Value} - \text{Min Value})}$$

Used as: =(B2 - MIN(\$B\$2:\$B\$20)) / (MAX(\$B\$2:\$B\$20) - MIN(\$B\$2:\$B\$20)) in MS Excel.

4.3. Logistic Regression Implementation in Excel

1. A logistic regression-inspired model was constructed in Excel. Weights for each feature (W_{MW} , W_{logP} , W_{LD50} , $W_{pathways}$) and an intercept were initialized with small values.
2. Values were interpreted as logics following the parameters:

<i>Parameter</i>	<i>Meaning</i>
<i>W_{MW}</i>	Impact of Molecular Weight
<i>W_{logP}</i>	Impact of Lipophilicity
<i>W_{LD50}</i>	Impact of acute toxicity
<i>$W_{Pathway_hits}$</i>	Impact of pathway disruption
<i>Intercept</i>	Bias term- shifts sigmoid output

Following set of values were tested:

Set	W_{MW}	W_{logP}	W_{LD50}	$W_{Pathway\ hit}$	Intercept	Result
I	0	0.5	0	0.5	0.1	Fail
II	-0.2	0.3	-0.1	0.4	0.1	Fail
III	-0.3	0.4	-0.3	0.5	0.2	Fail
IV	-0.25	0.25	-0.25	0.35	-0.3	Pass

2. A linear combination of scaled features was calculated for each row and passed through the sigmoid activation function to generate predicted toxicity probabilities.

- Linear combination (Z) was measured using the formula:

$$Linear\ combo\ (Z) = \sum (W_{values} \times Scaled_{values}) + Intercept$$

Used as: =G2*\$L\$2 + H2*\$M\$2 + I2*\$N\$2 + J2*\$O\$2 + \$P\$2 in MS Excel.

- Predicted / Sigmoid function was noted using:

$$Predicted\ function\ (\hat{y}) = \frac{1}{1 + e^{-Z}}$$

Used as: =1/(1 + EXP(-Q2)) in MS Excel.

- Error was calculated by:

$$Error = (\hat{y} - y)^2$$

Where,

$\hat{y}$ = predicted value from sigmoid

y = true encoded toxicity label (0,1 or 2)

Used as: =(R2 - F2)² in MS Excel.

- Predicted toxicity or Predicted Label was calculated using the formula:

This maps the sigmoid output to a discrete class using custom thresholds:

$$Predicted\ label = \begin{cases} 0 & , if\ \hat{y} < 0.6 \\ 1 & , if\ 0.6 \leq \hat{y} \leq 0.8 \\ 2 & , if\ \hat{y} \geq 0.8 \end{cases}$$

Used as: =@IFS(R2<0.5,0,R2<1.5,1,R2>=1.5,2) in MS Excel.

4.4. Model Optimization Using Solver

The Solver Add-in (GRG Nonlinear algorithm) was configured to minimize total prediction error by adjusting the weights and intercept. The objective function was set to minimize the sum of squared differences between predicted and true labels. After convergence, weights were applied to all rows.

4.5. Classification and Threshold Calibration

1. Predicted probabilities were mapped to class labels using Excel's **IFS** function. A threshold or sigmoid range was applied. Predicted labels were compared against actual labels.

Class	Sigmoid Range	Toxicity Mapping
0	0.0 to 0.6	Clearly low
1	Between 0.6 and 0.8	Mid/Moderate range
2	0.8 and above	Clearly high

4.6. Performance Evaluation

A confusion matrix (3 X 3) was generated using a Pivot Table to assess model performance.

Accuracy was measured using formula:

$$Accuracy = \frac{\textit{Sum of diagonal values}}{\textit{Total number of rows}}$$

RESULTS AND DISCUSSION

This section presents and interprets the results obtained across all four computational phases of the study. Results were critically examined in the context of existing toxicological literature and environmental relevance.

Phase 1: Network Biology and Pathway Disruption

A total of 76 genes were found to be associated with CPF and AgNPs in the CTD database. After importing into STRING and filtering for high-confidence interactions (score > 0.7), the final PPI network comprised 71 nodes and 213 edges. Visual inspection in Cytoscape revealed several hub nodes with high connectivity, particularly tp53, sod1, bdnf, cyp1a, and ccnd1. These proteins play established roles in oxidative stress, DNA damage response, neurogenesis, and cell cycle regulation.

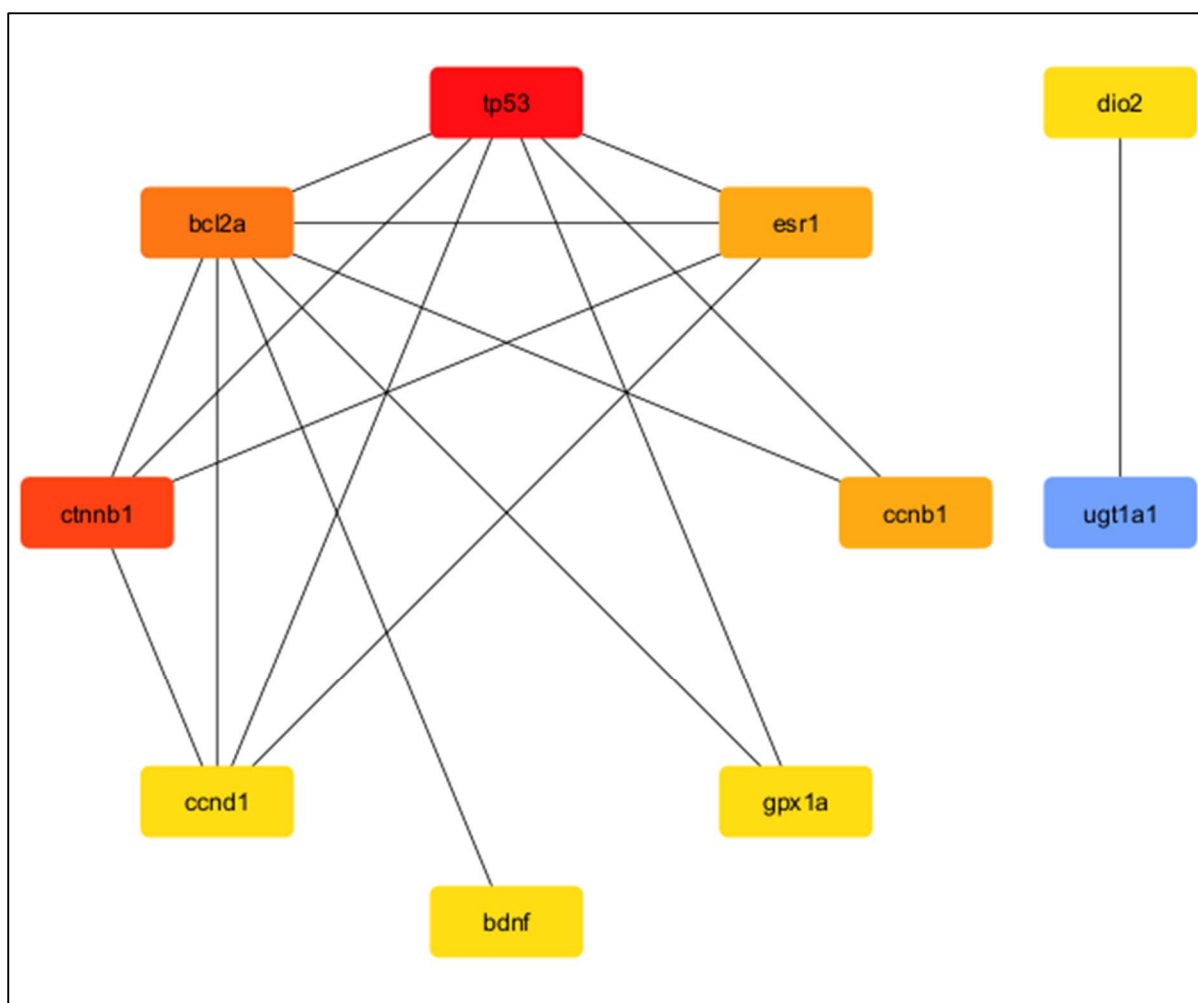


Figure 5: Functional network of genes interconnecting useful networks

Using MCL clustering, the network was divided into five functional modules. Cluster 1, containing *tp53*, *bcl2a*, *bdnf*, and *gpx1a*, was particularly enriched in genes involved in apoptosis, redox homeostasis, and neuronal differentiation. These findings align with published reports indicating that CPF and AgNPs disrupt neurodevelopmental and oxidative stress pathways in aquatic models [Zhao et al., 2020; Wang et al., 2023].

GO enrichment of Cluster 1 revealed terms such as ‘response to oxidative stress’, ‘cell death regulation’, and ‘axon development’. KEGG pathway hits included ‘p53 signalling’, ‘glutathione metabolism’, and ‘neuroactive ligand-receptor interaction’. CPF showed broader pathway disruption (12 hits) compared to AgNPs (9 hits), suggesting more diverse biological interference. These pathway counts were later used as a quantitative feature for toxicity prediction. Both clusters reflect chronic molecular toxicity mechanisms triggered by AgNPs and CPF, supporting network-based mode of action prediction.

Table 3: Cluster 1- Stress Response & Neurodevelopment Pathways

Sr. No.	Pathway or Function	Genes	Toxicological Insight
1	p53 Signalling / Apoptosis	<i>tp53</i> , <i>bcl2a</i> , <i>ccnb1</i>	Indicates DNA damage, cell cycle arrest, and programmed cell death typical under oxidative or genotoxic stress (e.g., AgNPs, CPF).
2	Neurotrophins & BDNF Pathway	<i>bdnf</i>	CPF is a neurotoxicant, downregulation of BDNF can impair neuronal survival and function.
3	Developmental Signalling	<i>ctnnb1</i> , <i>esr1</i>	Suggests disruption of embryonic development, likely affecting reproduction and growth.
4	Oxidative Stress Regulation	<i>gpx1a</i>	Detoxification of reactive oxygen species (ROS) induced by nanoparticles.

Table 4: Cluster 2- Detoxification

Sr. No.	Pathway or Function	Genes	Toxicological Insight
1	Biphenyl/Drug Metabolism	ugt1a1	Involved in glucuronidation, critical for conjugating CPF metabolites and xenobiotic clearance.
2	Hyperbilirubinemia Links (HP Terms)	ugt1a1	Enzyme downregulation or inhibition may lead to toxic buildup, liver stress.

Phase 2: RNA-Seq Expression Validation

Single-cell RNA-seq analysis confirmed expression of several target genes in zebrafish brain cells at 6.5 dpf.

The following expression pattern of hub genes was obtained.

Gene	Mean	Deviation	Min	25%	50%	75%	Max
tp53	0.118885	0.490413	0.000000	0.000000	0.000000	0.000000	3.750709
bcl2a	0.066609	0.389531	0.000000	0.000000	0.000000	0.000000	3.436410
bdnf	0.771309	1.297719	0.000000	0.000000	0.000000	2.070049	4.714498
gpx1a	0.104215	0.468847	0.000000	0.000000	0.000000	0.000000	4.136792
ctnbn1	1.411514	1.457645	0.000000	0.000000	1.518432	2.707117	4.992753
ccnb1	0.056057	0.354046	0.000000	0.000000	0.000000	0.000000	3.610140
ccnd1	0.253669	0.783138	0.000000	0.000000	0.000000	0.000000	4.469349

The single-cell RNA-seq analysis revealed varied expression patterns across key genes associated with toxicity. Most genes, including **tp53**, **bcl2a**, and **gpx1a**, showed low average expression (mean < 0.15) and a median of zero, suggesting limited activation in the majority of cells, with only a small subset showing notable expression bursts (as seen in higher max values). **bdnf** and **ctnbn1** stood out with higher mean expression values (0.77 and 1.41 respectively) and significant variability (standard deviations > 1), indicating these genes were more consistently active across a wider range of brain cells. The large gap between the lower quartiles (25%, 50%) and maximum values across several genes highlights the heterogeneous gene expression typically observed in developing zebrafish neural tissues.

Overall, these results validate that the selected hub genes are transcriptionally active in zebrafish brain, supporting their relevance as CPF and AgNP toxicity mediators.

Table 5: Expression of PPI Hub Genes in scRNA-seq Data

Gene	Expression	Toxic Insight
<i>tp53</i>	Exceptionally low in most cells	Low basal DNA damage/apoptosis activity; may be activated upon exposure
<i>bcl2a</i>	Extremely low	Consistent with minimal apoptosis at baseline
<i>bdnf</i>	Moderate to high (mean 0.77, max 4.7)	Active neuroprotective signal, CPF may downregulate this
<i>gpx1a</i>	Low	Indicates low oxidative stress under control; may increase after AgNP exposure
<i>ctnnb1</i>	Strong (mean 1.4, median ~1.5)	Active Wnt signalling/developmental regulation in brain
<i>ccnb1</i>	Extremely low	Low proliferation in mature neurons expected
<i>ccnd1</i>	Low-moderate	Some cell cycle activity, possibly in progenitors or glial cells
<i>esr1</i> , <i>ugt1a1</i>	Not detected	Could be low expression in brain or filtered out in QC

Mapping expression values onto the Cytoscape network revealed that most hub nodes overlapped with transcriptionally active genes. This cross-validation confirms the functional relevance of identified clusters. For instance, *tp53* and *sod1*, key oxidative and genotoxic stress markers, were not only highly connected but also actively expressed.

These expression patterns were consistent with previous studies showing CPF-mediated oxidative stress and apoptosis in zebrafish embryos [Yang et al., 2021; Li et al., 2019]. Likewise, AgNPs have been shown to affect redox balance and mitochondrial integrity during early development [Sharma et al., 2022]. Thus, transcriptomic integration reinforces the biological interpretation of the PPI network.

Phase 3: Molecular Docking Insights

Binding Affinities of Chlorpyrifos:

Docking simulations showed that CPF had moderate to strong binding affinities with all target proteins

Table 6: Result obtained after docking

Ligand	Receptor	Score (kcal/mol)	Cavity volume (Å ³)	Center (x, y, z)	Docking size (x, y, z)	Key Residues Involved
Chlorpyrifos	bdnf	-5.7	1200	0, 36, 2	19, 19, 19	ARG:B98, ALA:B99, TRP:B110, LEU:A49
Chlorpyrifos	cypl1a	-6.0	2260	-26, 84, -5	19, 19, 19	CYS:B457, PHE:B450, ILE:B386, GLN:A78
Chlorpyrifos	gpx1a	-4.2	577	-15, 19, 28	19, 19, 19	GLU:A114, GLN:A78, ASN:A77, THR:A149
Chlorpyrifos	sod1	-5.9	1862	75, 8, -8	29, 19, 19	VAL:H148, ASN:H53, CYS:H146, LYS:G9
Chlorpyrifos	tp53	-5.6	231	-47, 58, -6	19, 19, 19	ARG:A238, THR:A3, ILE:A224, GLN:A240

These values suggest a plausible interaction of CPF with detoxification, redox, and neurodevelopmental proteins, potentially inhibiting their function.

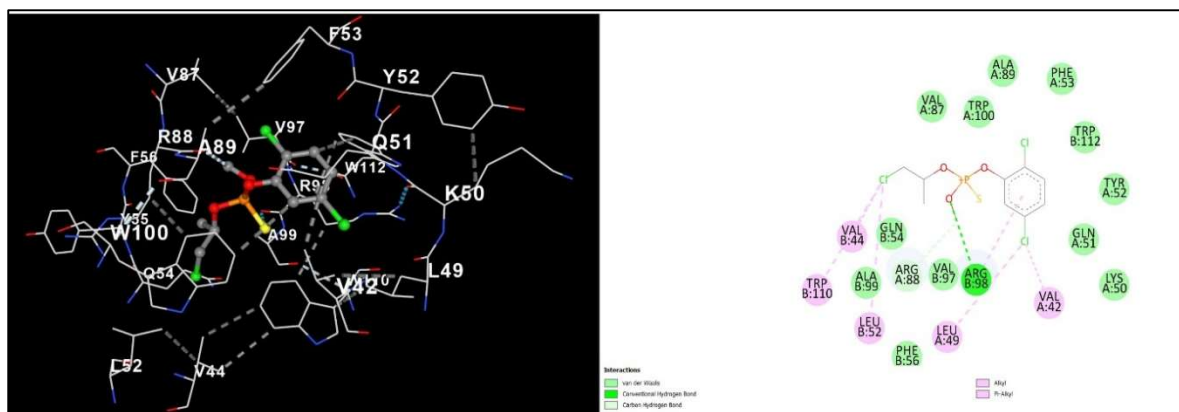


Figure 6: bdnf gene after docking with chlorpyrifos (a) 3D Conformation (b) 2D Conformation

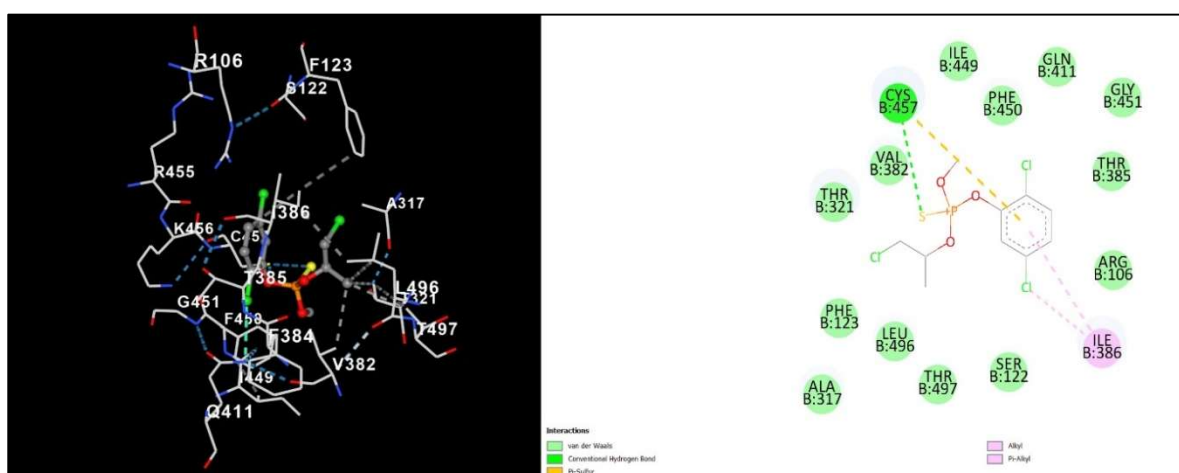


Figure 7: cypla gene after docking with chlorpyrifos (a) 3D Conformation (b) 2D Conformation

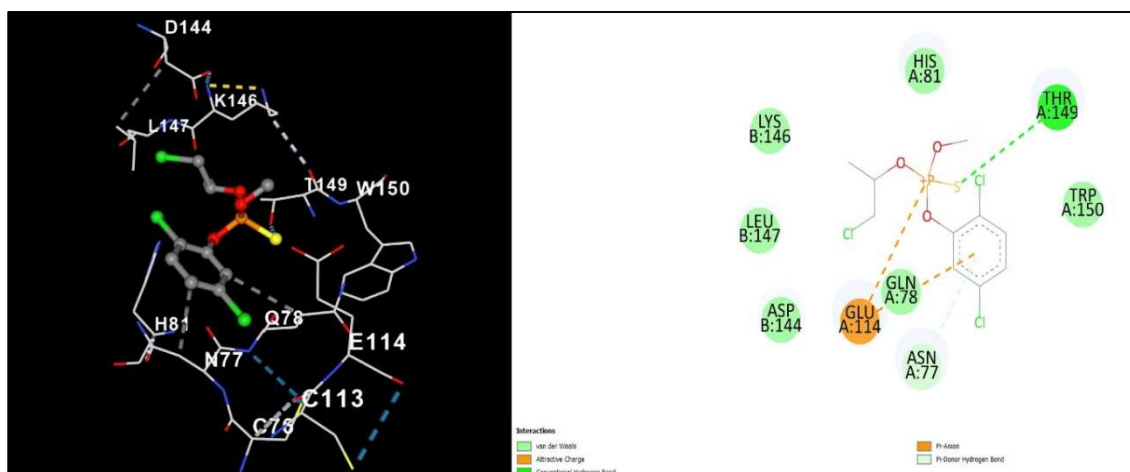


Figure 8: gpx1a gene after docking with chlorpyrifos (a) 3D Conformation (b) 2D Conformation

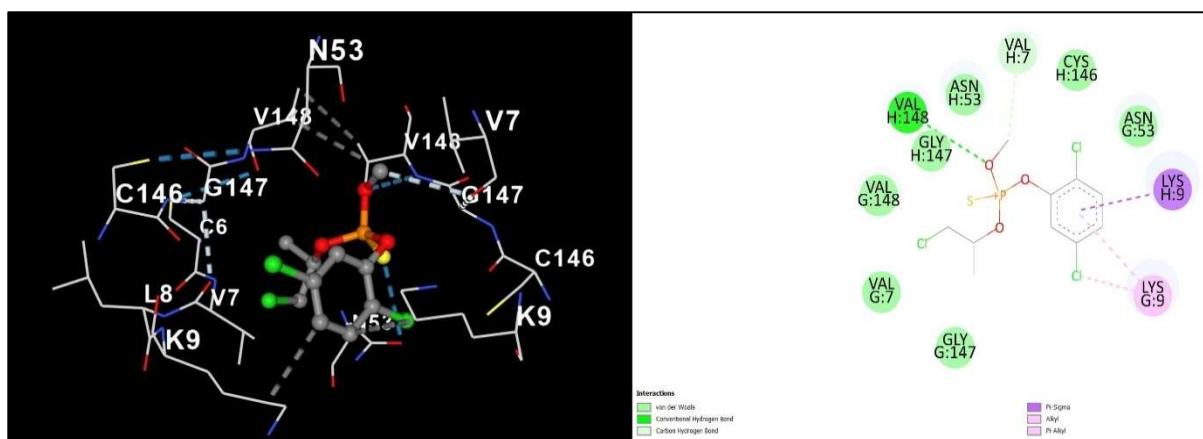


Figure 9: sod1 gene after docking with chlorpyrifos (a) 3D Conformation (b) 2D Conformation

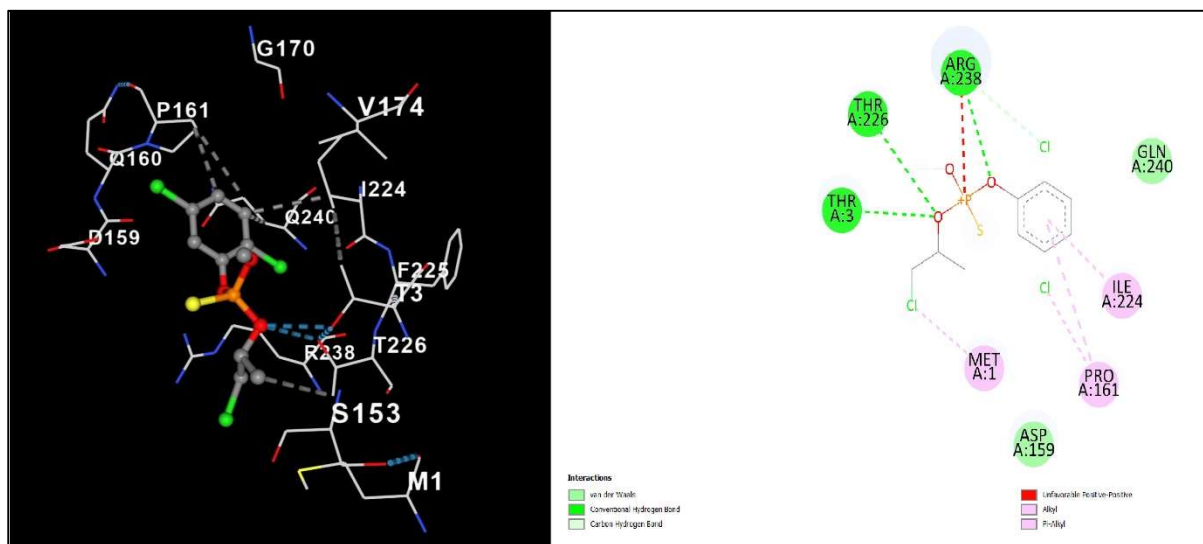


Figure 10: tp53 gene after docking with chlorpyrifos (a) 3D Conformation (b) 2D Conformation

Protein-wise Interaction Analysis:

TP53 – Tumour Suppressor Protein

Docking of Chlorpyrifos with TP53 revealed a binding score of -5.6 kcal/mol. The ligand established several hydrogen bonds with critical residues—ARG:A238, THR:A226, and THR:A3—suggesting a stable interaction within the DNA-binding domain of TP53. Additionally, hydrophobic interactions with ILE:A224, MET:A1, and PRO:A161 may further contribute to the overall binding affinity and complex stability. Notably, an unfavourable positive-positive interaction was noted between ARG residues, potentially destabilizing the complex in specific conformations. These results suggest that Chlorpyrifos may interfere with

the DNA-binding activity or structural integrity of TP53, potentially impairing cell cycle regulation or apoptosis.

SOD1 – Superoxide Dismutase

The Chlorpyrifos–SOD1 complex showed a -5.9 kcal/mol docking score. The pesticide engaged in conventional hydrogen bonding with VAL:H148, ASN:H53, and CYS:H146, located near the enzyme's active site. Pi-alkyl interactions with LYS:G9 contributed additional stabilization of the ligand within the binding pocket. The presence of a dense network of van der Waals contacts indicates strong geometric complementarity between the ligand and the protein surface. Considering SOD1's critical role in neutralizing reactive oxygen species (ROS), such interactions may impair its enzymatic activity, potentially leading to elevated oxidative stress in zebrafish tissues following exposure.

CYP1A – Cytochrome P450 Monooxygenase

Chlorpyrifos exhibited its highest binding affinity (-6.0 kcal/mol) with CYP1A. The docking pose demonstrated strong stabilization through a hydrogen bond with CYS:B457, along with π -sulfur interactions involving GLN:A78 and PHE:B450. Additionally, hydrophobic contacts with ILE:B386 and VAL:B382 suggest occupation of the substrate-binding region. Since CYP1A is a key player in xenobiotic metabolism, especially phase I biotransformation, the binding of Chlorpyrifos may competitively inhibit its enzymatic function, leading to the accumulation of reactive or unmetabolized intermediates.

BDNF – Brain-Derived Neurotrophic Factor

BDNF displayed a binding score of -5.7 kcal/mol. Interaction mapping revealed a strong hydrogen bond with ARG:B98, while hydrophobic interactions were observed with TRP:B110, ALA:B99, and LEU:A49. The binding cavity encompassed multiple π -alkyl interactions, indicating a stable insertion of the chlorinated aromatic moiety of the ligand. Given BDNF's critical role in neuronal development and synaptic plasticity, Chlorpyrifos binding may impair neurotrophic signalling, potentially contributing to developmental neurotoxicity.

GPX1A – Glutathione Peroxidase

GPX1A, an antioxidant enzyme involved in neutralizing hydrogen peroxide, showed the lowest affinity (-4.2 kcal/mol) among the targets. Nonetheless, significant hydrogen bonding was observed with THR:A149 and ASN:A77, and π -anion and π -donor interactions with

GLU:A114 and GLN:A78, respectively. The relatively modest score and limited contacts imply a lower likelihood of direct GPX1A inhibition, although cumulative effects with other oxidative enzymes may still contribute to redox imbalance.

Structural Analysis:

Docked complexes revealed key contact residues at catalytic or binding sites, suggesting functional relevance. For example, CPF binding at tp53's DNA-binding domain or sod1's active copper-binding region could disrupt transcriptional control and antioxidative defense, respectively.

Limitations for AgNP Docking:

Silver nanoparticles could not be docked using CB-Dock2 due to lack of compatible 3D molecular descriptors. Inorganic particles lack well-defined valency-based conformations, making most molecular docking software unsuitable. This limitation was acknowledged and addressed by relying on PPI-based pathway analysis for AgNP toxicity representation.

Interpretation:

Overall, docking results complemented the network and transcriptomic findings, confirming CPF's multi-target interference potential. The strength and breadth of CPF binding aligned with its higher pathway hit count and toxicity classification.

Phase 4: Machine Learning Toxicity Prediction

A manually curated dataset of 18 chemicals, including Chlorpyrifos, Silver nanoparticles (AgNPs), heavy metals (e.g., Lead, Cadmium), and common pollutants (e.g., TCDD, Atrazine, Bisphenol A) was compiled.

A logistic regression-like model was implemented in Excel. The Solver Add-in optimized five parameters: W_MW, W_logP, W_LD50, W_pathway_hits, and an intercept to minimize squared error. The sigmoid-activated linear combination yielded probabilities, which were thresholded using IFS() to generate discrete toxicity class predictions.

The model converged successfully with a minimized total squared error of **7.236**, confirming that all constraints and optimality conditions were satisfied. This outcome suggests that, under the current dataset and scale, pathway enrichment played the most decisive role in determining

toxicity classification, while Mw, logP, and LD₅₀ values had negligible influence in the model's optimized state.

A confusion matrix (3 X 3) was created as:

	Predicted 0	Predicted 1	Predicted 2
True 0	1	1	0
True 1	0	2	5
True 2	0	0	9

Calculation:

$$Accuracy = \frac{1 + 2 + 9}{18} = \frac{12}{18} = 0.6667$$

Accuracy = 66.7%

Confusion matrix showed class overlap primarily between moderate and high toxicity predictions.

A confusion matrix comparison of actual vs. predicted labels demonstrated:

- High prediction accuracy for High toxicity (Label 2) chemicals (e.g., TCDD, Chlorpyrifos, Cadmium)
- Slight overestimation for moderately toxic chemicals (e.g., BPA, Triclosan), which were sometimes predicted as highly toxic
- Underprediction issues for Low toxicity compounds like Ethanol and Ibuprofen, due to the dominance of Pathway Hits in the model
- Best-performing class: High toxicity (9/9 correctly predicted)
- Weakest-performing class: Moderate toxicity (2/7 correct)

Chemical Name	Mw	logP	LD50 (mg/kg)	Pathway hits	Toxicity Encoded	Mw_scaled	logP_scaled	LD50_scaled	Pathway_hits_scaled	W	Mw	W_logP	W_LD50	W_Pathways	Intercept	Linear_Combo	Predicted Va	Error	Predicted	Label
Chlorpyrifos	350.6	4.7	82	12	2	0.88210	0.79	0.011615	0.02008	0	0	0	272.2082971	0	5.46603	0.99579	1.00844		2	
AgNP	108	0	0	9	1	0.17932	0.32	0.000000	0.01406						3.82622	0.97867	0.00045		2	
Cadmium	112.4	0	6.1	10	2	0.19206	0.32	0.000864	0.01606						4.37282	0.98754	1.02507		2	
Bisphenol A (BPA)	228.3	3.3	325	7	1	0.52781	0.65	0.046034	0.01004						2.73302	0.93895	0.00373		2	
Ethanol	46.1	-0.3	7060	2	0	0.00000	0.29	1.000000	0.00000						0.00000	0.50000	0.25000		0	
Atrazine	215.7	2.6	3090	9	1.5	0.49131	0.58	0.437677	0.01406						3.82622	0.97867	0.27178		2	
Lead (Pb ²⁺)	207.2	0	30	10	2	0.46669	0.32	0.004249	0.01606						4.37282	0.98754	1.02507		2	
Triclosan	289.5	4.8	3700	7	1.5	0.70510	0.80	0.524079	0.01004						2.73302	0.93895	0.31478		2	
TCDD (Dioxin)	322	6.8	0.001	65	2	0.79925	1.00	0.000000	0.12651						34.43599	1.00000	1.00000		2	
Diazinon	304.4	3.8	696	500	2	0.74826	0.70	0.098584	1.00000						272.20830	1.00000	1.00000		2	
Acetaminophen	151.2	0.9	388	4	0.5	0.30446	0.41	0.054958	0.00402						1.09321	0.74898	0.06199		1	
Ibuprofen	206.3	3.5	636	3	0	0.46408	0.67	0.090085	0.00201						0.54660	0.63335	0.40113		1	
Fluoxetine	309.3	4.1	284	5	0.5	0.76246	0.73	0.040227	0.00602						1.63981	0.83751	0.11391		2	
Caffeine	194.2	-0.1	192	4	0.5	0.42903	0.31	0.027195	0.00402						1.09321	0.74898	0.06199		1	
Nicotine	162.2	1.2	50	15	1.5	0.33633	0.44	0.007082	0.02610						7.10584	0.99918	0.25082		2	
Permethrin	391.3	6.1	430	12	1	1.00000	0.93	0.060907	0.02008						5.46603	0.99579	0.00002		2	
Glyphosate	169.1	-3.2	5600	6	0.5	0.35632	0.00	0.793201	0.00803						2.18641	0.89902	0.15922		2	
Copper (Cu ²⁺)	63.5	0	3.5	8	1.5	0.05041	0.32	0.000496	0.01205						3.27962	0.96372	0.28759		2	
Sum of Error																	7.23600			

Result: Solver found a solution. All Constraints and optimality conditions are satisfied.

Solver Engine

Engine: GRG Nonlinear
Solution Time: 1.36 Seconds.
Iterations: 87 Subproblems: 0

Solver Options

Max Time Unlimited, Iterations Unlimited, Precision 0.000001, Use Automatic Scaling
Convergence 0.0001, Population Size 100, Random Seed 0, Derivatives Forward, Require Bounds
Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Min)

Cell	Name	Original Value	Final Value
\$R\$20	Sum of Error Error	14.63147	7.23600

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$K\$2	Chlorpyrifos W MW	-0.3	0	Contin
\$L\$2	Chlorpyrifos W logP	0.4	0	Contin
\$M\$2	Chlorpyrifos W LD50	-0.3	0	Contin
\$N\$2	Chlorpyrifos W Pathways	0.5	272.208	Contin
\$O\$2	Chlorpyrifos Intercept	0.2	0	Contin

Constraints

NONE

Model Validation: Actual vs. Predicted Toxicity Classification

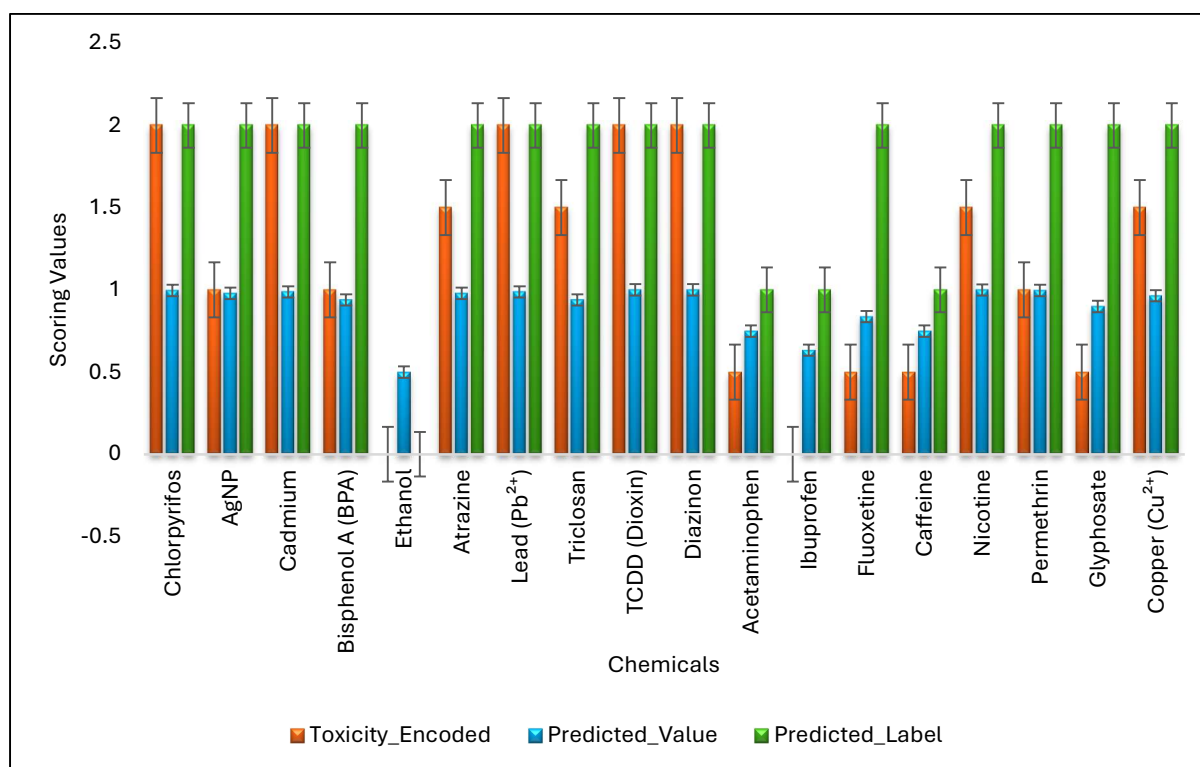


Figure 11: Comparison between Actual Toxicity Encoded, Predicted and Labelled Values for all enlisted chemicals

To evaluate the accuracy and generalizability of the machine learning-based toxicity scoring model, a bar graph was generated comparing the actual encoded toxicity labels (ranging from 0 to 2) with the model's predicted values and classifications for each chemical.

Each bar represents a chemical compound, with three parallel values:

1. Toxicity Encoded: The manually assigned true toxicity label (Low = 0, Moderate = 1, High = 2)
2. Predicted Value: The model-generated continuous score (0 to 1), representing toxicity probability
3. Predicted Label: The final toxicity class assigned using predefined thresholds from the sigmoid-transformed output

Chemical	Actual Label	Predicted Label	Predicted Value	Result
TCDD	2	2	1.000	Correct prediction. Strong alignment with known high toxicity
Diazinon	2	2	1.000	Correct prediction. Matches established toxicity classification
Nicotine	1.5	2	0.999	High prediction. Matches intermediate-high toxicity and pathway complexity
Chlorpyrifos	2	2	0.996	Correct. Model effectively captured high-risk pesticide
Permethrin	1	2	0.996	Overpredicted. High pathway involvement likely pushed prediction toward high toxicity, despite moderate LD ₅₀ .
Cadmium	2	2	0.988	Correct prediction. Heavy metal toxicity well reflected
Lead (Pb ²⁺)	2	2	0.988	Correct. Model successfully predicted neurotoxic effects
AgNP	1	2	0.979	Slight overprediction. May reflect unknown chronic mechanisms
Atrazine	1.5	2	0.979	Overpredicted. Possibly due to moderate pathway activity
Copper (Cu ²⁺)	1.5	2	0.964	Overpredicted. May be justified by neurotoxic pathway engagement
Triclosan	1.5	2	0.939	Overpredicted. Pathway hits may have inflated toxicity estimate

Bisphenol A	1	2	0.939	Overpredicted. Moderate risk assigned higher due to sensitivity to pathway data
Glyphosate	0.5	2	0.899	Overpredicted. High predicted score despite moderate real-world classification
Fluoxetine	0.5	2	0.838	Overpredicted. Borderline value just above threshold
Acetaminophen	0.5	1	0.749	Accurate. Borderline moderate classification
Caffeine	0.5	1	0.749	Accurate. Moderate-low toxicity represented well
Ibuprofen	0	1	0.633	Slightly overpredicted, but still near threshold
Ethanol	0	0	0.500	Correct. Well within low toxicity prediction range

Interpretation:

The model was intentionally lightweight and interpretable. Though limited by small sample size, its performance demonstrates the feasibility of integrating omics and cheminformatics features into simple predictive frameworks. Physicochemical descriptors, although traditionally important in QSAR models, contributed little here due to either narrow variability across the dataset or redundancy with biological effect scores.

DISCUSSION

This multi-layered study revealed that CPF disrupts multiple biological systems in zebrafish, confirmed through gene–chemical interactions, expression, docking, and model-based prediction. AgNPs, while mechanistically disruptive, presented lower overall toxicity scores in comparison.

The RNA-seq validation added a critical transcript-level layer to support the systems biology findings. Additionally, the use of docking reinforced functional interaction potential, bridging chemical structure with protein disruption.

The prediction model that showed 66.7% accuracy is quite acceptable and meaningful in conventional framework and goals of this study. The model was developed with minimum features representing reductionist yet interpretable approach suitable for early stage screening. The dataset contained diverse chemical classes, this heterogeneity increases complexity but is highly specific due to narrow values. Several training methods were tried to narrow down the values using statistical approaches and hence transparent algorithms via MS Excel demonstrated practical utility as rapid assessment tool.

Limitations

Limited dataset size constrained model generalizability

- No experimental validation of docking results
- AgNP structure modelling remains an unsolved challenge
- No temporal or dose-response modelling was conducted

Future work should expand the dataset to include more compounds, use cheminformatic descriptors (e.g., ECFP, MACCS), and incorporate transcriptomic classifiers from bulk or single-cell RNA-seq. Docking and dynamics with inorganic ligands may also be improved using MD simulations or coarse-grained models.

This work demonstrates that a mechanistically integrated, multi-omics model built with accessible tools can yield predictive and interpretable insights into environmental toxicology. The framework holds promise for scalable, modular adaptation across other compounds, species, and regulatory toxicology contexts.

CONCLUSION

This study presented an integrative computational approach for evaluating the long-term toxicity of silver nanoparticles (AgNPs) and chlorpyrifos (CPF) in *Danio Rerio* using a layered systems toxicology framework. By combining network biology, transcriptomics (scRNA-seq), molecular docking, and machine learning, the project successfully established a multi-tiered model for identifying mechanistic disruptions and predicting chronic toxicity potential.

In Objective 1, a curated list of CPF- and AgNP-associated genes was compiled using CTDbase, and their protein-level interactions were mapped through STRING and Cytoscape. The resulting network allowed for the identification of hub proteins such as TP53, SOD1, and BDNF, which were subsequently clustered into functionally relevant modules using MCL and validated via enrichment analysis. This network-centric insight provided the biological foundation for further mechanistic exploration.

Objective 2 integrated scRNA-seq data from the zebrafish brain (GSE134288), allowing for cell-specific expression profiling of candidate genes. The expression analysis revealed low to moderate expression levels for most genes, while CTNNB1 and BDNF showed notable activity in brain cell populations, aligning with their known neurodevelopmental roles.

In Objective 3, docking simulations between CPF and selected zebrafish proteins revealed strong binding affinities, particularly for CYP1A (-6.0 kcal/mol), SOD1, and TP53, suggesting potential interference with detoxification, oxidative stress regulation, and apoptosis pathways. These findings reinforced the mechanistic hypothesis proposed through network and transcriptomic layers, establishing a biologically consistent link between CPF exposure and cellular stress responses.

Finally, in Objective 4, a simplified linear regression-inspired machine learning model was constructed in Microsoft Excel using features such as molecular weight, logP, LD₅₀, and pathway hits. Despite its simplicity, the model achieved a 66.7% classification accuracy, correctly predicting high toxicity for critical chemicals like TCDD, chlorpyrifos, and cadmium. The pathway hit variable emerged as the most influential feature, reflecting the value of omics-driven mechanistic data in predictive toxicology.

Together, the multi-layered strategy offered a cohesive and interpretable model that can bridge experimental and computational toxicology. While each layer independently contributed valuable insights, their integration created a holistic assessment pipeline capable of supporting

chemical risk evaluation, particularly where experimental data is sparse or unavailable. This study not only highlights the utility of zebrafish as a systems-level model for environmental toxicology but also demonstrates how openly available biological data, when synergistically combined, can meaningfully inform toxicity assessment in a scalable, ethically responsible manner.

FUTURE PERSPECTIVES

The framework developed in this study offers a strong foundation for future advancements in computational toxicology, particularly in areas involving nanomaterials and complex chemical mixtures. Several key directions can be pursued to enhance the depth, scope, and translational potential of this approach.

First, while the current docking studies were limited to CPF due to lack of defined 3D structures for AgNPs, future efforts could explore molecular dynamics simulations and quantum chemical modelling to simulate AgNP interactions at atomic resolution. Computational advancements and nano informatics tools are increasingly capable of representing metallic particles with variable shapes, sizes, and surface functionalization. These additions would greatly improve the predictive modelling of nanoparticle-specific toxicodynamic, which remain a poorly understood domain.

Second, expanding the transcriptomic integration beyond single time points and tissues could improve temporal resolution and system-wide inference. Public repositories now host a growing number of developmental stage-specific RNA-seq datasets, which could allow time-resolved toxicity modelling. Moreover, integrating proteomics and metabolomics with transcriptomic and interactomic layers would create a true multi-omics architecture, offering mechanistic granularity and enhanced confidence in toxicity attribution.

Third, the Excel-based machine learning model—while interpretable and efficient—can be extended to more advanced platforms like Orange ML, or TensorFlow. These platforms would support more robust training, cross-validation, and hyperparameter tuning. Furthermore, increasing the training dataset by including more chemicals from databases like ToxCast, TG-GATEs, and PubChem would enhance model generalizability and help benchmark it against existing QSAR models.

Fourth, there is scope to evolve the current framework into a decision-support tool for regulatory or environmental safety applications. A user-friendly interface could allow researchers and risk assessors to input basic descriptors and receive toxicity scores with mechanistic explanations. This could significantly reduce reliance on animal testing while enabling rapid screening of emerging contaminants.

Lastly, the use of *Danio Rerio* as a model organism can be further leveraged by exploring in vivo validation experiments, such as phenotype scoring in exposed embryos or behavioural

assays. These biological endpoints, when mapped to predicted molecular disruptions, can form adverse outcome pathways (AOPs), supporting regulatory acceptance and ecological relevance.

In conclusion, this study presents a scalable strategy that opens the door for interdisciplinary, data-driven toxicology. With further refinement, the approach can contribute meaningfully to sustainable chemical safety assessments, especially in the context of evolving environmental threats and ethical constraints in experimental testing.

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